

NK cells for cancer: what makes a good product

Advances, failure modes and design decisions, across research and process-development evidence

313 curated records, 1993–2026 · 50 clinical trials · 58 process/manufacturing methods · 138 preclinical/mechanistic · 39 reviews · 244 with culture/manufacturing content

The product thesis: what is the best NK product we can build, and why

This collection (313 papers, 1993–2026) is read for one purpose: **to specify the best NK-cell cancer product that current evidence supports, and to say why each element is in it and what it displaces.** Everything that does not change a design decision is background.

The core argument in six sentences

NK-cell therapy's safety profile is settled — no GvHD, essentially no CRS/ICANS across allogeneic, cord-blood, iPSC and cell-line products (PMID 32023374, PMID 38238616, PMID 39798981, PMID 40251398, PMID 37148198). What is not settled is **potency-duration**: infused NK cells kill well for days and then stop mattering. Every advancement that has actually moved a clinical needle addresses potency-duration rather than target recognition — cell-intrinsic IL-15, memory-like programming, lymphodepletion with Treg control, protected CD16. Conversely, the dominant failure modes are not antigen escape (the CAR-T failure mode) but **metabolic collapse, TME reprogramming, CD16 shedding, trogocytosis-driven fratricide, and cryopreservation damage** — all of which are product-manufacturing problems as much as biology problems. Therefore the best product is not "the most sophisticated CAR" but **the most fitness-protected effector, manufactured in a process that does not destroy the fitness it was engineered for.** The rest of this note is that spec.

Recommendation: two products, not one

The evidence supports different answers for "build now" and "build for the platform".

Product A — near-term, highest probability of clinical benefit

Cord-blood-derived, feeder-expanded, memory-like-programmed NK cells, cryopreserved, dosed repeatedly with an IL-15-containing engager, after lymphodepletion.

No CAR. Specificity comes from the engager (TriKE/AFM13-class), which also supplies IL-15.

Why this and not a CAR product: - Every element has **clinical** support: CB expansion (PMID 24204673), CIML programming (PMID 27655849, PMID 35349491), engager pre-complexing in patients refractory to both brentuximab and checkpoint blockade (PMID 40186077), IL-15-in-the-engager for persistence (PMID 26847056), repeat dosing (PMID 40251398), Treg-depleting lymphodepletion tripling in vivo expansion (PMID 24719405). - It removes the three highest-risk failure modes of CAR-NK in one stroke: no transduction step (no VCN, clonality or vector-supply constraints), no CAR-driven trogocytosis/fratricide (PMID 36175679), and target swapping requires a new reagent, not a new product. - Manufacturing is short and cheap: 10–14 days, one banked CB unit yields many doses, engager loaded at formulation, and the complex survives freezing (PMID 35225870).

Cost: it depends on CD16, so it inherits shedding (mitigated with an ADAM17 inhibitor, PMID 39053944) and it will not solve solid tumours.

Product B — the platform worth building, 3-5 year horizon

iPSC-derived NK cells, clonal master bank, with:

Module	Purpose	Evidence
NK-native CAR (NKG2D-TM + 2B4 + CD3ζ), empirically costim-screened	antigen specificity that actually signals in NK cells	PMID 30082067; CD28/LCK surprise PMID 38900051
IL-15/IL-15Rα fusion (membrane-tethered or inducible)	persistence without systemic cytokine	PMID 39798981, PMID 28725044, PMID 32384544
hnCD16 (high-affinity, non-cleavable)	ADCC + serial killing + engager compatibility	PMID 31856277, PMID 41116262
CISH KO	cytokine-signalling brake; IL-15 independence	PMID 32531207, PMID 32902645
KLRC1/NKG2A KO	resist HLA-E inhibition	PMID 39349459, PMID 37675109
TGF-β resistance (SMAD4 KO or dnTGFBR2)	block TME reprogramming and NK→ILC1 conversion	PMID 40119192, PMID 38986609, PMID 28759001
Target-antigen KO where shared with NK (e.g. CD38)	prevent fratricide	PMID 32603414, PMID 33375774
Terminal IL-12/15/18 preactivation	memory-like state at the moment of infusion	PMID 22983442, PMID 27655849

Why iPSC rather than cord blood for the platform: **six or more edits plus a CAR is not a per-donor process**. Clonal selection, sequencing verification and one-time safety characterisation are only economic on a master bank, and multiplex-edited iNK products already exist clinically (PMID 39798981). Cord blood remains the better substrate for one- or two-modification products.

Why not NK-92 for either: irradiation before infusion removes persistence, which is the exact variable that determines efficacy. NK-92 remains a good vehicle for locoregional dosing (intracranial HER2-CAR NK-92, PMID 37148198) and for cheap preclinical/CMC development.

The five levers, ranked by evidence per unit of engineering cost

- 1. Cell-intrinsic IL-15 (or IL-15 in the engager).** Present in essentially every NK product with a clear clinical response signal. Highest return, well-characterised risk (unregulated IL-15 → GvHD signal after T-depleted HSCT, PMID 25452614; use tethered/inducible designs).
- 2. Host conditioning: lymphodepletion + Treg control.** Free — it is a regimen, not a product change — and it moved in vivo NK expansion from 10% to 27% of patients with improved AML clearance (PMID 24719405).
- 3. Memory-like (IL-12/15/18) programming.** 12-16 hours appended to an existing process; epigenetically stable (PMID 38941480); clinical expansion/persistence data in two settings.

4. **Protected CD16 (hnCD16 or ADAM17 blockade).** One module; converts single-round killers into serial killers (PMID 41116262, PMID 39666369) and makes every antibody/engager combination stronger.
5. **Brake knockouts (CISH → NKG2A → TGF-β).** Strongest preclinical case for solid tumours, immature clinical readout, and each edit adds safety-characterisation burden (PMID 41425600). Worth it on a clonal bank, hard to justify per-donor.

What I would cut

- **Novel CAR targets as the primary research investment.** Antigen choice is not the bottleneck; two independent solid-tumour CAR-NK reviews converge on trafficking, persistence and TME suppression instead (PMID 38245520, PMID 39134804), and CAR-NK relapse in vivo was metabolic, not antigenic (PMID 37494448).
- **Systemic IL-2 support.** It expands Tregs; IL-15 super-agonist or cell-intrinsic IL-15 dominates it.
- **Elaborate logic gates before fitness is solved.** synNotch-programmed TIGIT/CD73 conversion (PMID 38429294) is elegant, but a logic-gated cell that dies in 72 hours is still a cell that dies in 72 hours.
- **Fresh-only products.** They cannot be a commercial off-the-shelf product; the correct response to cryopreservation damage is to fix the formulation and the potency assay (`03_failure_modes.md`), not to avoid freezing.
- **Autologous NK products for most indications.** Patient NK cells are functionally impaired (PMID 20849361, PMID 40019438), which forfeits the platform's main advantage.

Reading order

- `01_scientific_advances.md` — the advances that earn a place in the spec, and the ones that don't yet.
- `02_manufacturing_advances.md` — the process advances, with numbers, and the unresolved platform comparison.
- `03_failure_modes.md` — ranked failure modes, mechanism, and the mitigation each implies.
- `04_product_spec_and_release.md` — the buildable spec, step by step, with the release panel.
- `05_evidence_base_and_open_questions.md` — clinical anchors, what is not established, and what would change this thesis.
- `REPORT.md` — the full 313-paper listing by axis (auto-generated).

What actually makes an NK product good: the decisions that matter

This collection (313 papers, 1993–2026) is read from a product point of view: **which advances change a design decision, which failure modes destroy products, and where the evidence genuinely supports a choice versus where it only supports an option**. Papers that describe biology without a design consequence stay in `REPORT.md`.

There is no single best NK product — the right design depends on target tissue, whether the antigen is shared with NK cells, how many modifications you can afford to characterise, and whether you are optimising for speed to clinic or for platform economics. What is well supported is a short list of levers, a shorter list of failure modes, and a set of trade-offs that are now well enough characterised to be made deliberately rather than by default.

The central claim

NK therapy's unsolved problem is potency-duration, not target recognition. Safety is settled: no GvHD and essentially no CRS/ICANS across allogeneic, cord-blood, iPSC-derived and cell-line products (PMID 32023374, PMID 38238616, PMID 39798981, PMID 40251398, PMID 37148198). But infused NK cells kill well for days and then stop mattering.

Two lines of evidence make this concrete rather than rhetorical:

- Preclinical CAR-NK relapse was driven by **loss of metabolic fitness, not antigen escape**, and was rescued by cytokine engineering (PMID 37494448).
- Cryopreserved NK cells pass conventional degranulation and release assays while losing **3-D migration and killing** (PMID 33067467) — so a product can meet specification and still be functionally inert in tissue.

Everything that has moved a clinical needle addresses potency-duration (cell-intrinsic IL-15, memory-like programming, lymphodepletion with Treg control, protected CD16); most of what limits products is a manufacturing-coupled fitness problem (metabolic collapse, TME reprogramming, CD16 shedding, trogocytosis-driven fratricide, cryo damage). That is the lens for the rest of these notes.

The levers, ranked by evidence per unit of engineering cost

1. **Cell-intrinsic IL-15, or IL-15 built into the engager.** Present in essentially every NK product with a clear clinical response signal (PMID 28725044 → PMID 32023374, PMID 38238616; PMID 39798981; PMID 26847056). Known risk: unregulated IL-15 was associated with acute GvHD after T-depleted HSCT (PMID 25452614), so tethered, fused or inducible designs exist for a reason (PMID 32384544).
2. **Host conditioning.** Free — a regimen, not a product change. Lymphodepletion enables expansion at all (PMID 15632206); adding Treg depletion moved in vivo NK expansion from 10% to 27% of patients with improved AML clearance (PMID 24719405).
3. **Memory-like (IL-12+IL-15+IL-18) programming.** 12–16 h appended to an existing process, epigenetically encoded rather than transient (PMID 22983442, PMID

38941480), with clinical expansion and persistence data in two settings (PMID 27655849, PMID 35349491).

4. **Protected CD16 (hnCD16 or ADAM17 blockade).** One module that converts single-round killers into serial killers (PMID 41116262, PMID 39666369) and strengthens every antibody or engager combination (PMID 31856277, PMID 39053944).
5. **Brake knockouts, in order of evidence: CISH → KLRC1/NKG2A → TGF-β resistance.** The strongest preclinical case for solid tumours; clinical readouts immature; each edit adds safety-characterisation burden (PMID 41425600), which is why edit count is an economic decision as much as a biological one.

The six decisions worth arguing about

Each is treated properly in 04_design_decisions.md; in short:

Decision	Options that survive scrutiny	What tips it
Cell source	cord blood; iPSC; peripheral blood; NK-92 for locoregional	number of modifications you must characterise, and doses per lot
Specificity	IL-15-containing engager vs CAR vs both	retargeting flexibility and transduction burden vs antigen-density control
Persistence	armour the cell vs put IL-15 in the engager vs support the host	how much regulatory control you want over IL-15 exposure
Edit count	0-2 (per-donor products) vs 4-6 (clonal bank)	translocation/clonality characterisation cost, not editing efficiency
Expansion	feeder cells vs cell-free particles vs feeder-free media	yield versus residual-cell testing and GMP feeder-bank burden
Cryopreservation	DMSO standard vs DMSO-minimised vs post-thaw rescue	whether your potency assay can even detect the damage

Where I would not spend the next dollar

- **Novel CAR targets as the primary research investment.** Two independent solid-tumour CAR-NK reviews converge on trafficking, persistence and TME suppression as the limiting factors (PMID 38245520, PMID 39134804), and in vivo relapse was metabolic rather than antigenic (PMID 37494448).
- **Systemic IL-2 support**, which expands Tregs — IL-15 super-agonist or cell-intrinsic IL-15 dominates it.
- **Elaborate logic gates before fitness is solved.** synNotch conversion of TIGIT/CD73 into activating inputs (PMID 38429294) is excellent science, but a logic-gated cell that dies in 72 hours still dies in 72 hours.
- **Fresh-only products.** The right response to cryo damage is a better formulation and a better potency assay (03_failure_modes.md, 05_potency_and_release.md), not avoiding the freeze that off-the-shelf logistics require.
- **Autologous NK products** for most indications: patient NK cells are functionally impaired (PMID 20849361, PMID 40019438), forfeiting the platform's main advantage.

Reading order

- `01_scientific_advances.md` — advances tiered by whether they change a decision.
- `02_manufacturing_advances.md` — process advances with numbers, and the platform comparison nobody has run.
- `03_failure_modes.md` — ranked failure modes, mechanisms, and the mitigation each implies.
- `04_design_decisions.md` — the six decisions, with the case for each option and when it wins.
- `05_potency_and_release.md` — why standard release testing misses the thing that matters, and what to measure.
- `06_clinical_evidence_and_open_questions.md` — trial-by-trial anchors, and what would change this thesis.

The scientific advances that change a design decision — and the ones that don't yet

Criterion for inclusion: **does this change a choice you would make when designing a product?** Papers that describe biology without a design consequence are in `REPORT.md`, not here.

Tier 1 — change the decision, with clinical support

1. Cell-intrinsic IL-15 (the highest-value single module)

IL-15 armoring converts a transient infusion into a product with measurable persistence. Secreted IL-15 + CD19 CAR in cord-blood NK cells (Liu 2018, PMID 28725044) became the clinical CAR19/IL-15 product (PMID 32023374; 37 patients with response determinants, PMID 38238616); an IL-15/IL-15R α **fusion** is a core module of iPSC-derived FT596 (PMID 39798981); soluble IL-15 defines cord-blood TRACK NK now in NSCLC (PMID 38572955, PMID 39903538).

Design consequence: the cytokine goes in the product, not in the patient. Tether it, fuse it or make it inducible (rimiducid-controlled MyD88/CD40 + IL-15, PMID 32384544) — the GvHD signal seen with IL-15/4-1BBL- activated NK cells after T-depleted HSCT (PMID 25452614) is the reason not to leave IL-15 unregulated.

2. Cytokine-induced memory-like programming (IL-12 + IL-15 + IL-18)

Romee 2012 (PMID 22983442) showed a 12–16 h preactivation produces weeks-long enhanced responsiveness; Romee 2016 (PMID 27655849) showed remissions in r/r AML; Berrien-Elliott 2022 (PMID 35349491) characterised expansion and persistence for post-transplant relapse. The state is **epigenetically encoded**, not a transient activation (PMID 38941480), which is why it survives manufacturing and infusion.

Design consequence: append it as the terminal step of any process. It costs one day, requires no genetic modification, and stacks with engagers (PMID 33986022), CARs (PMID 38996027), particle expansion (PMID 38711506) and checkpoint blockade (PMID 39948608).

3. Protected, high-affinity CD16

CD16A is shed by ADAM17 on activation — a negative-feedback loop that directly limits **serial killing** (PMID 41116262) and blunts engager and antibody combinations. Fixes: high-affinity non-cleavable CD16a in iPSC-NK (PMID 31856277, used in FT596), CD16-158V knocked into the CD38 locus so one edit both deletes CD38 and installs better CD16 (PMID 35135865), ADAM17 knockout (PMID 31704085), or pharmacological ADAM17 blockade which also enhances IL-15-driven activation and proliferation (PMID 39053944). The Fc γ RIIIA L48-H/R polymorphism enhancing ADCC via serial killing (PMID 39666369) tells you which variant to install.

4. IL-15-containing engagers (specificity without gene transfer)

The BiKE → TriKE progression is the field's most underrated result: CD16×CD33 BiKE established redirection (PMID 24652987), and adding IL-15 to make the 161533 TriKE added **expansion, persistence and survival** (PMID 26847056) — one reagent supplying both specificity and the Tier-1 lever above. Extended to CD19 in CLL (PMID 30890546), CLEC12A for AML stem cells (PMID 33097838), and NKp30×HER2 to bypass CD16 entirely (PMID 40694640). Trifunctional NKp46×CD16a×CD123 NKCE controlled AML and overcame CD64-mediated resistance to anti-CD123 ADCC (PMID 36635380).

AFM13 (CD30×CD16A) pre-complexed with cytokine-activated cord-blood NK cells gave "CAR-like" activity without gene transfer (PMID 33986022) and produced clinical responses in lymphoma refractory to both brentuximab and checkpoint inhibitors (PMID 40186077).

Design consequence: an engager moves specificity risk out of the cell product entirely — cheaper and retargetable — at the cost of depending on CD16 and on repeat dosing of a second agent.

5. Host conditioning is part of the product's efficacy

Miller 2005 (PMID 15632206) established that in vivo expansion requires lymphodepletion; Bachanova 2014 (PMID 24719405) showed IL-2-diphtheria-toxin Treg depletion raised the expansion rate from 10% to 27% of patients with better AML clearance. Donor selection is the other free lever: KIR B-haplotype donors improve survival after unrelated transplant for AML (PMID 20581313) and the HLA-B *21 M/T dimorphism shapes NK education and immunotherapy outcome (PMID 30647027).

Tier 2 — change the decision on preclinical strength, clinical readout pending

6. NK-native CAR architecture

Li 2018 (PMID 30082067) screened CARs in iPSC-NK cells: **NKG2D transmembrane + 2B4 + CD3ζ** outperformed the CD28-CD3ζ CAR-T design ported into NK cells. DAP10/DAP12-based receptors exploit native adaptors, and a KIR2D/DAP12 CAR reached the clinic in r/r B-ALL (PMID 34703879). But the design rule is screen, don't assume: CD28 — which mature NK cells do not express — augments a CD70-CAR via LCK/CD3ζ/ZAP70 (PMID 38900051), and knocking a ζ-less CAR into the endogenous CD3ζ locus gives potent cytotoxicity (PMID 38493479).

7. Brake knockouts

Ranked by evidence and by what each protects against. Which subset is worth taking depends on the edit-count decision in 04_design_decisions.md, not on biology alone:

Edit	Protects against	Key evidence
CISH	cytokine-signalling brake; IL-15 dependence	PMID 32531207; with IL-15-armoured CAR PMID 32902645; in vivo model PMID 35589278
KLRC1/NKG2A		

Edit	Protects against	Key evidence
	HLA-E-mediated inhibition, incl. CTC escape	PMID 39349459, PMID 37675109, PMID 35694192, PMID 36706761
SMAD4 / dnTGFB2	TGF- β + activin A reprogramming, NK $\rightarrow$ ILC1 conversion	PMID 40119192, PMID 38986609, PMID 34138753, PMID 28759001
CD38 (when using daratumumab or a CD38 CAR)	fratricide	PMID 32603414, PMID 33375774
ADAM17	CD16 shedding	PMID 31704085
ZC3H12A (Regnase-1), TIPE2, Cbl-b	intracellular restraint on IFN- γ / accumulation	PMID 38821052, PMID 36725083, PMID 24553136
HLA class I	host T-cell rejection of allogeneic NK	PMID 33584651

Unbiased screens are now the target pipeline: genome-wide CRISPR screens in primary human NK cells found MED12/ARIH2-class checkpoints whose ablation increases CAR-NK potency under immunosuppression (PMID 40845844), and in vivo AAV-SB screens of tumour-infiltrating NK cells across four solid-tumour models found genetic checkpoints of CAR-NK therapy (PMID 38918616). A different lever from the same technique: CHMP2A loss in glioblastoma stem cells and HNSCC makes the **target** NK-sensitive (PMID 35393416).

8. Metabolic fitness as a designed attribute

The most consequential mechanistic result for product design: single-cell profiling of CAR-NK cells after transfer showed relapse was driven by **loss of metabolic fitness, reversible by cytokine engineering** (PMID 37494448). The supporting map: amino-acid-dependent cMyc as the checkpoint (PMID 29904050), TGF- β repression of mTOR (PMID 26884601) and metabolic dysfunction in human metastatic breast cancer (PMID 33568351), hypoxia via HIF-1 α (PMID 32445619) and SHP-1/STAT3/ERK (PMID 33123602), obesity/PPAR-driven lipid paralysis (PMID 30420624), lactate-driven lysine **lactylation** that can be targeted to restore killing (PMID 40494934), vitamin B6 restriction by pancreatic cancer (PMID 37931287), myCAF-driven glutamine deprivation (PMID 39696364), and mitochondrial apoptotic priming setting killing efficiency at physiological E:T ratios (PMID 35447071).

Design consequence: treat metabolic fitness as a **release attribute** and a design target, not a research topic (see 05_potency_and_release.md).

Tier 3 — real advances that do not yet change a design decision

- **High-dimensional phenotype maps.** scRNA-seq/CITE-seq resolution of human NK states (PMID 38956378), pan-cancer tumour-infiltrating NK maps (PMID 38956379, PMID 37607536) and tissue imprinting (PMID 32059780) prove CD56^{bright/dim} is inadequate — but no map yet names a state you can select for and manufacture at scale. Their present use is **product characterisation**, and that is genuinely valuable: it is how you show your process makes the same cell twice.

- **Adaptive/NKG2C+ NK cells.** Associated with reduced leukaemia relapse (PMID 26416461) and now buildable into iPSC-NK products (PMID 34525347) — but it depends on CMV serostatus for primary products, so it is a donor-selection footnote until the engineered version has clinical data.
- **Synthetic-biology logic.** synNotch conversion of TIGIT/CD73 into activating signals (PMID 38429294) and KIR-based inhibitory CARs (PMID 36175679) are excellent science;
the iCAR is worth its complexity mainly where trogocytosis is demonstrated (see 03_failure_modes.md).
- **Checkpoint blockade of NK cells.** Monalizumab/anti-NKG2A unleashes NK and T cells (PMID 30503213) and TIGIT blockade prevents exhaustion (PMID 29915296) — but for a cell product, deleting the receptor beats co-administering an antibody.
- **Pharmacological sensitisation of the tumour.** Bortezomib lowering HLA-E and raising DR5 (PMID 30713790), BET inhibition via BRD4/SMAD3 (PMID 38519469), STING agonists with CAR-NK (PMID 35371622), radiotherapy via CXCL8 (PMID 35319989). These are combination-regimen decisions, not product decisions — but they are the cheapest way to improve a fixed product.
- **NK→DC→T-cell axis.** NK recruitment of cDC1 (PMID 29429633) and the NK-DC axis in checkpoint-responsive tumours (PMID 29942093) argue for positioning NK products with T-cell therapy rather than against it — a clinical-development choice.

The manufacturing advances that matter, with numbers

Dose and post-thaw fitness are the two manufacturing outputs that determine whether a good design becomes a good product. This note ranks the process advances by how much they change those two outputs.

1. Membrane-bound ligand presentation solved the dose problem (2009-2016)

The single most consequential manufacturing lineage in the field. Soluble cytokines cannot reproduce sustained, contact-dependent costimulation; engineered K562 feeders can.

Platform	Ligands	Reported yield	Source
K562-mb15-41BBL	membrane IL-15 + 4-1BBL	median 21.6-fold NK in 7 days (5.1–86.6, n=50)	Imai/Fujisaki 2009, PMID 19383914
K562-mb15-41BBL in G-Rex	+ gas-permeable static culture	up to 19 × 10⁹ NK cells in 8–10 days from unseparated apheresis	Lapteva 2012, PMID 22900959
K562-mbIL21-41BBL aAPC	membrane IL-21 + 4-1BBL	~1,848-fold (fresh CB) / ~2,389-fold (cryopreserved CB) in 14 days, >95% purity	Shah 2013, PMID 24204673
mbIL-21 feeder line ("NKF")	membrane IL-21	>10,000-fold at 5 weeks, with increased metabolic activation	PMID 31624330
K562-OX40L + short soluble IL-21	OX40L	robust expansion via OX40–OX40L	PMID 31105701
EBV-LCL feeders	—	clinical-grade cytotoxic NK, upregulated activating receptors and death-receptor ligands	PMID 19308771

Two things to take from the table. First, **IL-21 replaced IL-15 as the expansion ligand** because it buys proliferative capacity without terminal differentiation — and the mbIL-21 platform also raises metabolic activation (PMID 31624330), which is the attribute that later predicts in vivo durability (PMID 37494448). Second, the vessel is not a detail: the same feeder in a gas-permeable vessel produced a ~1,000× larger absolute yield (PMID 22900959).

Cost of feeders: an irradiated tumour-derived line inside the process means a GMP feeder bank, irradiation validation and residual-feeder-cell release testing.

2. Cell-free mimics of feeders (the advance that removes the feeder liability)

PM21 particles — plasma-membrane particles from K562-mb21-41BBL cells — give feeder-like expansion (mean ~825-fold) with in vivo biodistribution data (PMID 27059202), and with added cytokines yield memory-like characteristics plus improved survival from one

process (PMID 38711506). This is the most mature "feeder benefit without feeder cells" route and is the strongest candidate to make the feeder-versus-feeder-free argument moot.

Feeder-free alternatives worth knowing: automated feeder-free closed-system bioreactor expansion of clinical-grade NK cells (PMID 20795758), feeder-free expansion integrated with non-viral editing of **cryopreserved** primary NK cells (PMID 33433623), and a 2025 review categorising cytokine-only, immobilised-ligand and nanoparticle aAPC-mimic strategies against feeder-based yield (PMID 41089682).

3. Media and supplements are a potency variable, not a procurement detail

A 2025 Cytotherapy study screened six media (RPMI 1640, KBM581, SCGM, NK MACS, X-VIVO 15, AIM-V) against four supplements (FBS, human AB serum, human platelet lysate, Immune Cell Serum Replacement) with feeders, reading out viability, fold expansion, cytotoxicity, immunophenotype **and** transcriptome (PMID 39570247). Medium/supplement choice changed both yield and phenotype. The practical implication is unwelcome: expansion data are not portable between labs running different media, which is a large part of why the platform comparison in §6 is missing.

4. Automation and closure moved NK manufacturing from artisanal to process

- **CliniMACS Prodigy**: selection → activation → expansion → transduction in one closed device, used with autologous feeders + IL-21 for clinical-grade NK and anti-CD123 CAR-NK (PMID 28810809).
- **End-to-end automated primary CAR-NK manufacture** for AML has been demonstrated (PMID 38253870) — the relevant milestone is that CAR-NK production no longer requires an open, operator-dependent process.
- **Feeders raise transduction, not just proliferation** (PMID 35222382; engineered feeders displaying screened costimulatory molecules + IL-21 improve both, PMID 40325497). Delivery method and expansion platform therefore have to be co-optimised; choosing them independently leaves efficiency on the table.
- **Gene delivery** options now include baboon-envelope pseudotyped lentivirus (used in a clinical CD19-BBz CAR-NK product, PMID 40251398), alpharetroviral vs lentiviral head-to-head (PMID 32117200), mRNA/plasmid electroporation (PMID 31114587), Cas9 RNP electroporation reaching ~90% editing of primary NK (PMID 31704085), and site-specific knock-in that combines a knockout with a knock-in in one edit (PMID 35135865, PMID 38493479).

5. Scale-up is finally displacing scale-out

Flask/bag processes scale by multiplication; bioreactors scale by volume. Process development for NK expansion in **aerated stirred bioreactors** (PMID 40808775) plus recent scalable NK/CAR-NK bioreactor process work is the current frontier, and it is where the field is thinnest — almost all published yield data come from static vessels, and virtually none from edited or CAR-transduced cells in stirred systems.

For iPSC products the analogous advance is **clinical-scale, feeder-free differentiation** (PMID 23515118, PMID 31396935), with the caveat that differentiation strategy itself changes phenotype and cytotoxicity (PMID 39445004) — i.e. the differentiation protocol is a critical process parameter, not a preamble.

6. What the manufacturing literature still cannot tell you

- **No head-to-head platform comparison.** Nobody has run feeder cells vs PM21 particles vs feeder-free media on shared donors with one potency assay and one cryopreservation step. Every yield comparison across papers is confounded by medium, vessel, donor and assay.
- **Non-comparable potency reporting.** Expansion fold, E:T ratio, target line and assay format differ so widely that quantitative meta-analysis is not possible — a real limitation of this corpus, not just of this review.
- **Cryopreservation is under-reported in exactly the papers that need it.** Clinical papers rarely report post-thaw function, despite the evidence that it is where potency is lost (PMID 33067467) — see 03_failure_modes.md and 05_potency_and_release.md.
- **Cost of goods is essentially absent**, even though it is the central argument for allogeneic, cord-blood and iPSC approaches.

Failure modes, ranked

Ranked by how often they determine the outcome and by how expensive they are to discover late. The first four are the ones that have actually sunk products or explained disappointing trials; the rest are real but narrower.

1. Metabolic collapse after infusion (the dominant efficacy failure)

Mechanism. Effector function depends on glycolysis/OXPHOS with amino-acid-dependent cMyc as the checkpoint (PMID 29904050). The tumour removes exactly those inputs: hypoxia via HIF-1 α (PMID 32445619) and SHP-1-mediated attenuation of STAT3/ERK (PMID 33123602); TGF- β repression of mTOR (PMID 26884601) and metabolic dysfunction in human metastatic breast cancer (PMID 33568351); lactate-driven lysine **lactylation** of NK proteins (PMID 40494934); vitamin B6 restriction by pancreatic cancer (PMID 37931287); myCAF exosome-driven glutamine deprivation in colorectal liver metastasis (PMID 39696364); PPAR-driven lipid accumulation in obesity (PMID 30420624); C/EBP β -dependent autophagy inhibition (PMID 39609420). Mitochondrial apoptotic priming sets killing efficiency at physiological E:T ratios (PMID 35447071).

Why it is #1. In vivo single-cell profiling showed CAR-NK relapse coincided with metabolic collapse — not antigen loss — and cytokine engineering reversed it (PMID 37494448).

Mitigations, in order of evidence: cell-intrinsic IL-15; CISH knockout to raise cytokine-signalling gain (PMID 32531207, PMID 32902645); TGF- β resistance (PMID 40119192); expansion platforms that raise metabolic activation (mbIL-21, PMID 31624330); non-genetic levers such as DHA (PMID 38526128) or fasting-induced niche remodelling (PMID 38878769). And measure it — see failure mode #4.

2. Cryopreservation damage that release testing cannot see

Mechanism. Cryopreserved NK cells retain degranulation and 2-D chromium-release activity while losing **3-D migration and cytotoxicity in tissue-like matrices** (Mark 2020, PMID 33067467); fresh-versus-thawed comparison of clinical-grade expanded NK cells confirms functional loss (PMID 32536506). Recent mechanism: lysosomal damage triggering programmed cell death, alleviated by induced stress granules (PMID 42098076), and composition/temperature/granule-dependent post-thaw survival (PMID 42564981); a dedicated damage-mechanism study appeared in Cytotherapy (PMID 39918490).

Why it is #2. It is silent. A product can meet every release specification and be inert in tissue, and because clinical papers rarely report post-thaw function, its contribution to weak solid-tumour results is unquantified.

Mitigations: DMSO-minimised or DMSO-free formulations benchmarked properly (PMID 41208818, PMID 42098076, PMID 42564981); intracellular nanoparticle-mediated cryoprotection (PMID 32382476); post-thaw rescue by brief co-culture restoring motility and killing (PMID 40966444); and — non-negotiable — a **3-D post-thaw potency assay** (05_potency_and_release.md). Note that complex products can be frozen: engager-

precomplexed NK cells retain activity and specificity after cryopreservation (PMID 35225870).

3. CD16 shedding and the serial-killing ceiling

Mechanism. Activation triggers ADAM17-mediated CD16A cleavage, which limits serial killing (PMID 41116262) — the killing that matters at the low E:T ratios found in tumours. Antibody- and engager-dependent designs are hit hardest.

Mitigations: hnCD16 (PMID 31856277, clinically deployed in FT596, PMID 39798981); CD16-158V knock-in (PMID 35135865); ADAM17 knockout (PMID 31704085); ADAM17 inhibition, which additionally enhances IL-15-driven activation (PMID 39053944). Donor/variant choice matters too (FcγRIIIA L48-H/R, PMID 39666369).

4. Trogocytosis-driven fratricide and antigen dilution (CAR-specific, easy to miss)

Mechanism. CAR engagement transfers cognate antigen from tumour to NK cell. The tumour's antigen density falls and the antigen-decorated NK cells become targets for each other. Li 2022 (PMID 36175679) demonstrated both and rescued them with a **KIR-based inhibitory CAR**.

Related, simpler case: self-antigen fratricide. NK cells upregulate CD38 during expansion, so CD38-CAR NK cells kill each other (fixed by CD38 knockout plus affinity-tuned CAR, PMID 33375774) and daratumumab kills the product (fixed by CD38 knockout, PMID 32603414).

Design rule this implies: before committing to a CAR target, check whether the antigen is expressed on activated NK cells or transferable by trogocytosis. If yes, plan a knockout or an inhibitory CAR from the start — not after the in vivo experiment fails.

5. TME reprogramming of the infused product

Mechanism. TGF-β converts effector NK cells into ILC1-like cells (PMID 28759001), a lineage-level loss of function, and drives it through SMAD3/E4BP4 (PMID 28262747); CXCR3-driven infiltration produces ILC1-like plasticity in liver metastasis (PMID 40168086). Beyond TGF-β: Tregs suppress NK cells TGF-β-dependently (PMID 16230475), MDSCs via NKp30 in HCC (PMID 19551844), CAFs act as cytotoxicity decoys in breast cancer (PMID 40052789) and reshape NK phenotype in melanoma (PMID 19934056), αv-integrin/TGF-β limits activity against GBM stem cells (PMID 34138753). Infused NK cells can even become counterproductive — tumour-associated NK cells driving MDSC-mediated tolerance via IL-6/STAT3 (PMID 38748775). Most sobering: NK cells become functionally impaired **within hours** of entering a tumour (PMID 38267402).

Mitigations: TGF-β/activin resistance (SMAD4 knockout, PMID 40119192; required for iPSC-NK efficacy against HCC, PMID 38986609); NKG2A/KLRC1 deletion for HLA-E-high tumours (PMID 39349459, PMID 36706761); sensitising the tumour instead of the cell (bortezomib lowering HLA-E, PMID 30713790; BET inhibition, PMID 38519469; STING agonists, PMID 35371622; CHMP2A loss making GBM stem cells NK-sensitive, PMID

35393416); or bypassing trafficking altogether with locoregional delivery (intracranial CAR-NK-92, PMID 37148198).

6. Trafficking failure in solid tumours

Distinct from #5: cells that never arrive cannot be reprogrammed. Chemokine-receptor engineering is the direct answer — CCR7 by trogocytosis for lymph-node homing (PMID 22498742), CCR7 co-delivered with a CAR (PMID 31114587), CXCR3 in liver metastasis (PMID 40168086); homing biology is reviewed at PMID 35768424. The pragmatic answer, where anatomy allows, is intratumoral or intracavitary dosing (PMID 37148198).

7. Persistence ceiling, and the cost of raising it

Even armoured products persist weeks, not months. Raising it has a documented cost: IL-15/4-1BBL-activated NK cells were associated with **acute GvHD** after T-cell-depleted HSCT (PMID 25452614) — the clearest reminder that cytokine armouring is not free. Repeat dosing (PMID 40251398) is the pragmatic mitigation; inducible cytokine circuits (PMID 32384544) are the engineered one.

8. Manufacturing-induced failures that are entirely self-inflicted

- **Feeder residuals and irradiation validation** — a controllable but real CMC burden; PM21 particles (PMID 27059202) or feeder-free processes (PMID 41089682) remove it.
- **Medium/supplement switching silently changing potency and phenotype** (PMID 39570247).
- **Editing-related genomic risk outrunning its characterisation.** Multiplex editing raises translocation and clonality questions; safety-assessment frameworks for CRISPR-modified NK cells are only now being formalised (PMID 41425600). This — not editing efficiency — is what limits edit count.
- **Irradiated cell lines cannot persist by construction.** NK-92 products are safe (PMID 18836917, PMID 24094496) but the irradiation that makes them safe removes the variable that drives efficacy.
- **Autologous starting material that is already impaired:** patient NK cells show diminished cytotoxicity (PMID 20849361) and marrow NK deficits predict worse daratumumab response in myeloma (PMID 40019438).

The pattern

Six of the eight failure modes are **fitness** failures rather than recognition failures, and four of them are at least partly created or fixed in manufacturing. That is the strongest argument in this collection for treating process development and biology as one problem.

The six design decisions, and the case for each option

Each decision below has more than one defensible answer. What the literature supports is a clear statement of what you gain, what you accept, and which conditions tip the choice.

1. Cell source

Option	Strongest case	What you accept	Tips toward it
Cord blood	Banked, HLA-typed, expands ~1,800–2,400-fold with aAPC feeders (PMID 24204673); substrate for the IL-15-armoured CD19 CAR-NK clinical programme (PMID 28725044 → PMID 32023374, PMID 38238616) and TRACK NK (PMID 38572955)	Immature phenotype (CD56bright-skewed, low KIR) needing priming; still a per-lot process	1–2 modifications, fastest credible route to clinic
iPSC	Clonal bank makes multiplex editing economic; clinical proof with FT596 (CD19 CAR + hnCD16 + IL-15/IL-15Rα, PMID 39798981); feeder-free clinical-scale differentiation (PMID 31396935, PMID 23515118)	Differentiation protocol is a critical process parameter that changes phenotype and cytotoxicity (PMID 39445004); tumourigenicity and genomic characterisation burden	≥3 edits, off-the-shelf economics, uniformity as a selling point
Peripheral blood	Mature, educated, immediately cytotoxic; enables KIR/HLA donor selection (PMID 20581313, PMID 30647027); the historical clinical base (PMID 15632206, PMID 21791425)	Donor variability, low starting frequency, T-cell depletion required	Donor-selection-driven strategies, transplant settings, academic trials
NK-92 / cell lines	Unlimited, consistent, cheap; safe at high doses (PMID 18836917, PMID 24094496); CAR versions clinically dosed intracranially (PMID 25373520 → PMID 37148198)	Irradiation before infusion removes persistence entirely	Locoregional dosing, frequent redosing, CMC and preclinical development

Autologous products are the weakest option in most indications: patient NK cells are functionally impaired (PMID 20849361, PMID 40019438).

2. Where specificity comes from: engager, CAR, or both

Engager (IL-15-containing). Specificity and persistence in one reagent (161533 TriKE, PMID 26847056); retargeting means a new reagent, not a new product; no transduction step, so no VCN/clonality burden; clinical responses in double-refractory CD30+ lymphoma with AFM13-precomplexed cord-blood NK cells (PMID 40186077); NKp46-directed constructs escape CD16 dependence and CD64-mediated resistance (PMID 36635380, PMID 40694640). Accept: CD16 dependence (failure mode #3) and a second agent to dose.

CAR. Direct control of antigen-density thresholds and signal strength, and the only route to targets with no Fc-engager equivalent. Use NK-native architecture (NKG2D-TM + 2B4 + CD3 ζ , PMID 30082067) but screen costimulation empirically — CD28 works in NK cells despite not being expressed there (PMID 38900051), and CD3 ζ -locus knock-in is a viable alternative topology (PMID 38493479). Accept: transduction step, trogocytosis/fratricide risk (PMID 36175679), one target per product.

Both. CAR plus a cytokine-activated/memory-like backbone is already clinical (CAR-CIML NK against mesothelin, PMID 38996027), and CAR-T cells secreting BiKEs to recruit endogenous NK cells (PMID 40659834) show the field expects effector classes to be combined rather than ranked.

3. How persistence is supplied

Three routes, and they are substitutes, not complements:

- **In the cell** — secreted IL-15 (PMID 28725044), IL-15/IL-15R α fusion (PMID 39798981), membrane/intracellular IL-15 (PMID 22310931), or inducible MyD88/CD40 + IL-15 (PMID 32384544). Most persistence per dose; least regulatory control over exposure (GvHD signal, PMID 25452614).
- **In the engager** — IL-15 as a module of the TriKE (PMID 26847056). Exposure stops when dosing stops, which is the cleanest safety story; requires the engager to be part of the regimen forever.
- **In the host** — lymphodepletion plus Treg control (10% $\rightarrow$ 27% in vivo expansion, PMID 24719405), or IL-15 super-agonist support (used with CIML NK plus CTLA-4 blockade in head and neck cancer, PMID 39948608). Cheapest, and it works on any product — but systemic IL-2 specifically should be avoided because it expands Tregs.

Pairing CISH knockout with any of these lowers the cytokine concentration needed (PMID 32531207, PMID 32902645).

4. How many edits to take

This is an economics decision, not a biology decision: Cas9 RNP electroporation already reaches ~90% editing of primary NK cells (PMID 31704085) and feeder activation improves it further (PMID 35222382). The constraint is that multiplex editing raises translocation and clonality questions whose assessment frameworks are only now being formalised (PMID 41425600).

Priority order if you can take n edits:

1. CISH — cytokine-signalling gain, persistence, IL-15 independence (PMID 32531207).
2. Protected CD16 — hnCD16 knock-in or ADAM17 knockout (PMID 31856277, PMID 31704085); can be combined with a CD38 knockout in a single CD38-locus knock-in (PMID 35135865).
3. KLRC1/NKG2A — for HLA-E-high tumours (PMID 39349459, PMID 37675109).
4. TGF- β resistance — SMAD4 knockout or dnTGFBR2, for solid tumours specifically (PMID 40119192, PMID 38986609).

5. Target-antigen knockout where the antigen is shared with NK cells (CD38 is the worked example, PMID 32603414).
6. Emerging, screen-derived: Regnase-1/ZC3H12A (PMID 38821052), TIPE2 (PMID 36725083), Cbl-b (PMID 24553136), MED12/ARIH2-class hits (PMID 40845844, PMID 38918616).

For a per-donor product, 1–2 edits is the realistic ceiling; the 4–6 edit stack is what justifies a clonal iPSC bank in the first place.

5. Expansion platform

Option	Case	Accept
Engineered feeder cells (mbIL-21/4-1BBL)	Highest, best-replicated yields (PMID 19383914, PMID 24204673, PMID 31624330); also improves transduction (PMID 35222382, PMID 40325497)	GMP feeder bank, irradiation validation, residual-cell release testing
Cell-free particles (PM21)	Feeder-like expansion (~825-fold) without feeder cells, with in vivo data (PMID 27059202); with cytokines gives memory-like phenotype and better survival (PMID 38711506)	Particle manufacture and characterisation become your problem instead
Feeder-free media/ligands	Simplest CMC; closed automated feeder-free expansion is established (PMID 20795758) and integrates with non-viral editing of cryopreserved cells (PMID 33433623)	Generally lower yield; strategy landscape still consolidating (PMID 41089682)

Vessel and automation are separable choices layered on top: gas-permeable static culture for absolute yield (19×10^9 cells, PMID 22900959), CliniMACS Prodigy for closure and integrated transduction (PMID 28810809, PMID 38253870), stirred bioreactors for true scale-up (PMID 40808775) — the last being the least evidence-backed for edited or CAR-transduced cells. Media/supplement choice is a potency variable in its own right (PMID 39570247).

6. Cryopreservation strategy

Standard DMSO is the default and it demonstrably damages tissue-relevant function (PMID 33067467, PMID 32536506). The options are not "freeze or not" — an off-the-shelf product must freeze — but:

- **DMSO-minimised / DMSO-free formulations**, benchmarked against CryoStor 10 and FBS+DMSO (PMID 41208818, PMID 42098076, PMID 42564981).
- **Intracellular nanoparticle protection**, which changes the cell rather than the buffer (PMID 32382476).
- **Post-thaw rescue** — brief co-culture restoring motility and killing after thaw (PMID 40966444).
- **Freeze the complex** where relevant: engager-precomplexed NK cells retain activity and specificity (PMID 35225870), which is what makes engager-based off-the-shelf logistics viable.

Whichever you choose, the decision is only meaningful if the potency assay can detect the damage — which is the subject of 05_potency_and_release.md.

Two configurations these decisions produce

Not prescriptions — worked examples of coherent decision sets:

- **Fast, low-risk:** cord-blood NK, feeder- or PM21-expanded, terminal IL-12/15/18 programming, no edits, cryopreserved DMSO-minimised, dosed repeatedly with an IL-15-containing engager after Treg-depleting lymphodepletion. Every element has clinical support; specificity risk sits in a reagent, not the cell.
- **Platform play:** iPSC-derived NK, NK-native CAR + IL-15/IL-15R α + hnCD16, CISH/KLRC1/TGF- β -resistance edits, feeder-free clinical-scale differentiation, terminal memory-like programming, released on a 3-D post-thaw potency assay. Higher characterisation burden, and the only configuration that plausibly addresses solid tumours.

Potency and release: why standard testing misses the thing that matters

This is the most actionable gap in the field. If one change had to be made to how NK products are developed, it would be here.

The problem, stated precisely

Conventional NK potency readouts — chromium/calcein release, CD107a degranulation, IFN- γ secretion — are 2-D, short-duration, single-round and run at high E:T ratios. Cryopreserved NK cells **pass** them while having lost 3-D migration and cytotoxicity in tissue-like matrices (Mark 2020, PMID 33067467). The assay's blind spot lines up exactly with the conditions that matter in a patient: low E:T, tissue mechanics, repeated engagement.

Consequences visible in this corpus:

- Fresh-versus-thawed functional loss is measurable when investigators look for it (PMID 32536506) but is rarely reported in clinical papers, so its contribution to weak solid-tumour efficacy is unquantified.
- "NK activity" is not comparable across products or trials, which is a large part of why the manufacturing platform comparison in `02_manufacturing_advances.md` has never been run.

Assays that track the right thing

- **3-D spheroid / matrix cytotoxicity.** High-throughput 3-D spheroid microarrays for NK cytotoxicity (PMID 34290356); 3-D colorectal spheroid killing that was independent of target PD-L1 (PMID 29632716) — i.e. 3-D formats change conclusions, not just numbers.
- **Serial killing.** Reveals the CD16-shedding ceiling that single-round assays cannot see (PMID 41116262) and distinguishes CD16 variants (PMID 39666369).
- **Patient-derived organoid co-culture.** Shows hypoxia/TGF- β -driven, patient-specific diversification of NK activation programmes (PMID 41360426) — the closest available proxy for what the product will meet.
- **Post-thaw migration,** given that migration is the function cryopreservation destroys first (PMID 33067467).
- **Metabolic readouts.** Resazurin-based viability validated for NK-derived products (PMID 39022723); metabolic fitness is the attribute that predicted in vivo durability (PMID 37494448) and is raised or lowered by the expansion platform (PMID 31624330) and by freezing.
- **High-dimensional phenotyping of the final product.** Now reported alongside clinical outcomes (PMID 38238616, PMID 39948608), which is what makes phenotype-based release criteria (memory-like signature, CD16 level and cleavage status, exhaustion markers) credible rather than aspirational.

A defensible release panel

Attribute	Assay	Why it is here
Identity / purity	CD56+CD3– frequency; residual T cells; residual feeder cells	standard; feeder residuals are a real CMC risk (PMID 19383914-class processes)
Viability and recovery	post-thaw, both reported	viability alone hides cell loss
Potency (primary)	3-D or spheroid cytotoxicity, post-thaw, at clinically plausible E:T	the 2-D assay's documented blind spot (PMID 33067467)
Potency (secondary)	serial-killing capacity; CD107a/IFN- γ on receptor restimulation	catches CD16 shedding (PMID 41116262)
Fitness	post-thaw migration; metabolic readout	tracks the failure mode that drove in vivo relapse (PMID 37494448)
Phenotype	CD16 (and cleavage status), NKG2A/NKG2D/NCRs, memory-like and exhaustion markers	ties release to the state you intended to manufacture (PMID 22983442, PMID 38941480)
Genetic modification	VCN, transgene expression, editing efficiency, off-target and translocation assessment	edit count is limited by this, not by editing efficiency (PMID 41425600)
Safety	sterility, mycoplasma, endotoxin, irradiation controls for feeders	standard

What would have to be true for this to become standard

A cross-lab qualified 3-D or serial-killing potency assay with defined acceptance criteria. Nothing in this corpus suggests that exists yet, and its absence is the single largest obstacle to comparing products, comparing manufacturing platforms, or interpreting a disappointing trial as a biology failure versus a product failure.

Clinical evidence, and what would change the argument

What patients have actually shown

Safety is the settled claim. No GvHD and little or no CRS/ICANS across allogeneic, cord-blood, iPSC-derived, engineered and cell-line products, including CD19 CAR-NK (PMID 32023374, PMID 38238616), iPSC-derived FT596 (PMID 39798981), 4-1BB-costimulated CD19 CAR-NK (PMID 40251398) and intracranial HER2 CAR-NK-92 (PMID 37148198); the 2025 landscape review catalogues 120+ CAR-NK trials (PMID 40149002).

Efficacy signals cluster where three things coincide: a haematological target, an armoured or memory-like-programmed product, and real lymphodepletion.

Product	Setting	What it establishes
Haploidentical PB-NK + IL-2	AML, non-transplant	in vivo expansion and remissions require lymphodepletion (PMID 15632206); Treg depletion raised expansion 10%→27% with better clearance (PMID 24719405)
KIR-ligand-mismatched NK	elderly high-risk AML	feasible, no GvHD, CR in active disease (PMID 21791425)
CB-derived CAR19/IL-15 NK	r/r CD19+ B-cell malignancies	phase 1/2 (PMID 32023374), then 37 patients with determinants of response (PMID 38238616) — the reference dataset for armoured CAR-NK
CD19-BBz CAR-NK (BaEV-transduced CB)	r/r large B-cell lymphoma	repeat dosing is feasible and tolerable (PMID 40251398)
FT596 (iPSC; CD19 CAR + hnCD16 + IL-15/IL-15Rα)	B-cell lymphoma	a multiplex-engineered iPSC product can be dosed in humans (PMID 39798981)
Donor CIML NK	post-transplant relapse	memory-like programming expands and persists in patients (PMID 35349491; original AML trial PMID 27655849)
CIML NK + N-803 ± ipilimumab	r/r head and neck cancer	the most credible solid-tumour attempt: state programming + cytokine support + checkpoint blockade (PMID 39948608)
AFM13-precomplexed CB-NK	CD30+ lymphoma refractory to brentuximab and checkpoint inhibitors	engager-loaded cells can work without any gene transfer (PMID 40186077)
TRACK NK (CB, sIL-15, PD-L1+)	checkpoint-refractory NSCLC	interim engineered-NK solid-tumour data (PMID 39903538, PMID 38572955)
NK-92 / CAR-NK-92	RCC, melanoma, recurrent GBM	safe at high doses; locoregional delivery is viable (PMID 18836917, PMID 24094496, PMID 37148198)
Autologous SNK01 + chemo/cetuximab	NSCLC after TKI failure	autologous route is feasible but starts from impaired cells (PMID 38538093)

Solid tumours remain largely unsolved, and the two least-negative examples both added something beyond "more NK cells": intracranial delivery that bypasses trafficking (PMID 37148198), and combined state programming plus cytokine support plus checkpoint blockade (PMID 39948608).

Ranked by strength of clinical evidence

1. IL-15 armouring or IL-15-containing engagers — in essentially every product with a clear response signal.
2. Lymphodepletion plus Treg control — oldest, best-quantified, and free (PMID 24719405).
3. Memory-like programming — clinical expansion/persistence in two settings.
4. Donor/education selection — outcome associations in transplant cohorts (PMID 20581313, PMID 30647027).
5. Protected/high-affinity CD16 plus an antibody or engager — mechanistically strong, now inside clinical products.
6. Brake knockouts — compelling preclinically, clinical readouts immature.
7. Route of administration for solid tumours — one positive locoregional trial.

What is genuinely unresolved

- **No qualified potency assay**, so "the product met specification" does not imply potency, and cross-trial comparison of NK activity is unsound (05_potency_and_release.md).
- **Persistence still measured in weeks**. Cell-intrinsic IL-15 helps; nothing approaches CAR-T-like durability, and repeat dosing is the current pragmatic answer (PMID 40251398).
- **No head-to-head manufacturing comparison** on shared donors with one potency readout: feeder vs PM21 vs feeder-free, static vs bioreactor, DMSO vs DMSO-free.
- **Multi-edit safety science lags the edit lists** that screens are now generating (PMID 41425600 vs PMID 40845844, PMID 38918616).
- **Solid-tumour TME resistance** — hypoxia, TGF- β , lactylation, nutrient competition, CAF decoys, and functional impairment within hours of tumour entry (PMID 38267402) — is the bottleneck, and no clinical product yet carries the full resistance package.
- **Cost of goods is essentially unreported**, despite being the main argument for allogeneic and iPSC platforms.

What would change the argument in these notes

- A trial in which an unarmoured product matched an armoured one → IL-15 is not the dominant lever, and the ranking above collapses.
- A cryopreserved product showing preserved 3-D function with standard DMSO → failure mode #2 downgrades to a formulation detail.
- A solid-tumour response driven by a novel CAR target alone → antigen choice deserves more investment than argued in 00_thesis_and_decision_frame.md.

- Multi-month persistence from a cell-intrinsic circuit → the "repeat dosing" default is wrong, and edit-count economics change with it.
- A head-to-head platform comparison showing feeder-free yield parity → the feeder CMC burden becomes indefensible.

Honest limits of this collection

Axis, evidence class, disease tag and manufacturing/engineering flags are title/abstract-level heuristics, useful for navigation and not a substitute for reading. Citation counts are Europe PMC's and are time-biased against 2025-2026 records. Per-axis quotas mean the corpus is deliberately balanced across topics rather than proportional to publication volume — clinical and process evidence are over-weighted relative to the raw literature on purpose. Anything not resolvable from an abstract was checked against the 153 mirrored open-access full texts where available; the remaining records are catalogued from metadata only.

NK cells for cancer — evidence listing

313 curated records, 1993–2026. 244 (77%) report expansion, culture or manufacturing content; 153 have open-access full text mirrored under `fulltext/` (marked FT).

Evidence mix — Clinical trials: 50, Human translational: 28, Process / manufacturing methods: 58, Preclinical (in vivo): 105, Mechanistic / in vitro: 33, Reviews: 39

Disease context — pan-cancer: 149, lymphoma_cll: 32, aml_mds: 32, gi: 22, hnscc_melanoma: 21, myeloma: 18, breast: 14, gbm_cns: 12, neuroblastoma_sarcoma: 11, ovarian: 10, gu: 9, all_peds_heme: 8, lung: 6

Cross-cutting NK-cell therapy landscape

22 papers · 1994–2024 · 17 with culture/manufacturing content

Clinical trials

- **2002** Uherek C et al. — Retargeting of natural killer-cell cytolytic activity to ErbB2-expressing cancer cells results in efficient and selective tumor cell destruction. *Blood*. [PMID 12149207](#)
- **2006** Roda JM et al. — Natural killer cells produce T cell-recruiting chemokines in response to antibody-coated tumor cells. *Cancer research*. [PMID 16397268](#)
- **2009** Mustjoki S et al. — Clonal expansion of T/NK-cells during tyrosine kinase inhibitor dasatinib therapy. *Leukemia*. [PMID 19295545](#)

Human translational

- **2021** Mathewson ND et al. — Inhibitory CD161 receptor identified in glioma-infiltrating T cells by single-cell analysis. *Cell*. [PMID 33592174](#)
- **2023** Tang F et al. — A pan-cancer single-cell panorama of human natural killer cells. *Cell*. [PMID 37607536](#)

Process / manufacturing methods

- **1998** Vitale M et al. — NKp44, a novel triggering surface molecule specifically expressed by activated natural killer cells, is involved in non-major histocompatibility complex-restricted tumor cell lysis. *The Journal of experimental medicine*. [PMID 9625766](#) FT
- **1999** Cantoni C et al. — NKp44, a triggering receptor involved in tumor cell lysis by activated human natural killer cells, is a novel member of the immunoglobulin superfamily. *The Journal of experimental medicine*. [PMID 10049942](#) FT
- **2012** Kimura H et al. — EBV-associated T/NK-cell lymphoproliferative diseases in nonimmunocompromised hosts: prospective analysis of 108 cases. *Blood*. [PMID 22096243](#)

Preclinical (in vivo)

- **1994** Kägi D et al. — Cytotoxicity mediated by T cells and natural killer cells is greatly impaired in perforin-deficient mice. *Nature*. [PMID 8164737](#)

- **2001** Diefenbach A et al. — Rae1 and H60 ligands of the NKG2D receptor stimulate tumour immunity. *Nature*. [PMID 11557981](#)
- **2005** Gasser S et al. — The DNA damage pathway regulates innate immune system ligands of the NKG2D receptor. *Nature*. [PMID 15995699](#)
- **2005** Palumbo JS et al. — Platelets and fibrin(ogen) increase metastatic potential by impeding natural killer cell-mediated elimination of tumor cells. *Blood*. [PMID 15367435](#)
- **2009** Cella M et al. — A human natural killer cell subset provides an innate source of IL-22 for mucosal immunity. *Nature*. [PMID 18978771](#)
- **2024** Colucci M et al. — Retinoic acid receptor activation reprograms senescence response and enhances anti-tumor activity of natural killer cells. *Cancer cell*. [PMID 38428412](#) FT
- **2024** Kath J et al. — Integration of ζ-deficient CARs into the CD3ζ gene conveys potent cytotoxicity in T and NK cells. *Blood*. [PMID 38493479](#) FT
- **2024** Delconte RB et al. — Fasting reshapes tissue-specific niches to improve NK cell-mediated anti-tumor immunity. *Immunity*. [PMID 38878769](#)

Mechanistic / in vitro

- **2008** Caligiuri MA — Human natural killer cells. *Blood*. [PMID 18650461](#)
- **2024** Dean I et al. — Rapid functional impairment of natural killer cells following tumor entry limits anti-tumor immunity. *Nature communications*. [PMID 38267402](#) FT

Reviews

- **2008** Vivier E et al. — Functions of natural killer cells. *Nature immunology*. [PMID 18425107](#)
- **2016** Guillerey C et al. — Targeting natural killer cells in cancer immunotherapy. *Nature immunology*. [PMID 27540992](#)
- **2024** Vivier E et al. — Natural killer cell therapies. *Nature*. [PMID 38383621](#)
- **2024** Page A et al. — Development of NK cell-based cancer immunotherapies through receptor engineering. *Cellular & molecular immunology*. [PMID 38443448](#) FT

NK phenotypes & cell states (subsets, adaptive/memory-like, dysfunction)

26 papers · 2009–2024 · 15 with culture/manufacturing content

Clinical trials

- **2018** Guillaumón CF et al. — NK Cell Education in Tumor Immune Surveillance: DNAM-1/KIR Receptor Ratios as Predictive Biomarkers for Solid Tumor Outcome. *Cancer immunology research*. [PMID 30242020](#)
- **2019** Hallner A et al. — The HLA-B -21 dimorphism impacts on NK cell education and clinical outcome of immunotherapy in acute myeloid leukemia. *Blood*. [PMID 30647027](#)
- **2022** Shapiro RM et al. — Expansion, persistence, and efficacy of donor memory-like NK cells infused for posttransplant relapse. *The Journal of clinical investigation*. [PMID 35349491](#) FT

Human translational

- **2017** Pesce S et al. — Identification of a subset of human natural killer cells expressing high levels of programmed death 1: A phenotypic and functional characterization. The Journal of allergy and clinical immunology. PMID 27372564
- **2021** Shaim H et al. — Targeting the αv integrin/TGF- β axis improves natural killer cell function against glioblastoma stem cells. The Journal of clinical investigation. PMID 34138753
- **2024** Netskar H et al. — Pan-cancer profiling of tumor-infiltrating natural killer cells through transcriptional reference mapping. Nature immunology. PMID 38956379 FT

Process / manufacturing methods

- **2009** Balsamo M et al. — Melanoma-associated fibroblasts modulate NK cell phenotype and antitumor cytotoxicity. Proceedings of the National Academy of Sciences of the United States of America. PMID 19934056
- **2014** Ali TH et al. — Enrichment of CD56(dim)KIR + CD57 + highly cytotoxic NK cells in tumour-infiltrated lymph nodes of melanoma patients. Nature communications. PMID 25472612 FT
- **2016** Cichocki F et al. — CD56dimCD57+NKG2C+ NK cell expansion is associated with reduced leukemia relapse after reduced intensity HCT. Leukemia. PMID 26416461

Preclinical (in vivo)

- **2011** Platonova S et al. — Profound coordinated alterations of intratumoral NK cell phenotype and function in lung carcinoma. Cancer research. PMID 21708957
- **2012** Romee R et al. — Cytokine activation induces human memory-like NK cells. Blood. PMID 22983442
- **2017** Gao Y et al. — Tumor immunoevasion by the conversion of effector NK cells into type 1 innate lymphoid cells. Nature immunology. PMID 28759001
- **2017** Wagner JA et al. — CD56bright NK cells exhibit potent antitumor responses following IL-15 priming. The Journal of clinical investigation. PMID 28972539
- **2017** Sun C et al. — High NKG2A expression contributes to NK cell exhaustion and predicts a poor prognosis of patients with liver cancer. Oncoimmunology. PMID 28197391
- **2020** Ni J et al. — Single-Cell RNA Sequencing of Tumor-Infiltrating NK Cells Reveals that Inhibition of Transcription Factor HIF-1 α Unleashes NK Cell Activity. Immunity. PMID 32445619
- **2021** Woan KV et al. — Harnessing features of adaptive NK cells to generate iPSC-derived NK cells for enhanced immunotherapy. Cell stem cell. PMID 34525347
- **2024** Tarannum M et al. — CAR memory-like NK cells targeting the membrane proximal domain of mesothelin demonstrate promising activity in ovarian cancer. Science advances. PMID 38996027 FT
- **2024** Ding Y et al. — Distinct developmental pathways generate functionally distinct populations of natural killer cells. Nature immunology. PMID 38872000

Mechanistic / in vitro

- **2013** Horowitz A et al. — Genetic and environmental determinants of human NK cell diversity revealed by mass cytometry. Science translational medicine. PMID 24154599
- **2017** Freud AG et al. — The Broad Spectrum of Human Natural Killer Cell Diversity. Immunity. PMID 29166586
- **2020** Dogra P et al. — Tissue Determinants of Human NK Cell Development, Function, and Residence. Cell. PMID 32059780
- **2024** Rebuffet L et al. — High-dimensional single-cell analysis of human natural killer cell heterogeneity. Nature immunology. PMID 38956378 FT
- **2024** Hofman T et al. — IFN γ mediates the resistance of tumor cells to distinct NK cell subsets. Journal for immunotherapy of cancer. PMID 38955423 FT
- **2024** Foltz JA et al. — Cytokines drive the formation of memory-like NK cell subsets via epigenetic rewiring and transcriptional regulation. Science immunology. PMID 38941480

Reviews

- **2017** Bi J et al. — NK Cell Exhaustion. Frontiers in immunology. PMID 28702032 FT
- **2018** Chiossone L et al. — Natural killer cells and other innate lymphoid cells in cancer. Nature reviews. Immunology. PMID 30209347

Cell sources: peripheral blood, cord blood, iPSC, cell lines

20 papers · 1994–2026 · 18 with culture/manufacturing content

Clinical trials

- **2013** Tonn T et al. — Treatment of patients with advanced cancer with the natural killer cell line NK-92. Cytotherapy. PMID 24094496
- **2016** Suck G et al. — NK-92: an 'off-the-shelf therapeutic' for adoptive natural killer cell-based cancer immunotherapy. Cancer immunology, immunotherapy : CII. PMID 26559813
- **2017** Zhang C et al. — Chimeric Antigen Receptor-Engineered NK-92 Cells: An Off-the-Shelf Cellular Therapeutic for Targeted Elimination of Cancer Cells and Induction of Protective Antitumor Immunity. Frontiers in immunology. PMID 28572802 FT
- **2024** Lu T et al. — Preclinical Evaluation of Off-The-Shelf PD-L1+ Human Natural Killer Cells Secreting IL15 to Treat Non-Small Cell Lung Cancer. Cancer immunology research. PMID 38572955
- **2025** Villalona-Calero MA et al. — Interim report on engineered NK cell trial in lung cancer refractory to immune checkpoint inhibitors. JCI insight. PMID 39903538 FT

Human translational

- **2026** Zhang TY et al. — Study on the Mechanisms of Allogeneic NK Cells Combined with Myeloid Cells in Treating Acute Myeloid Leukemia. Transplantation and cellular therapy. PMID 42035877

Process / manufacturing methods

- **2012** Della Chiesa M et al. — Phenotypic and functional heterogeneity of human NK cells developing after umbilical cord blood transplantation: a role for human cytomegalovirus?. *Blood*. PMID 22096237
- **2013** Ribeiro A et al. — Mesenchymal stem cells from umbilical cord matrix, adipose tissue and bone marrow exhibit different capability to suppress peripheral blood B, natural killer and T cells. *Stem cell research & therapy*. PMID 24406104 FT
- **2016** Koehl U et al. — Advances in clinical NK cell studies: Donor selection, manufacturing and quality control. *Oncoimmunology*. PMID 27141397 FT
- **2024** Huyghe M et al. — Comparative analysis of iPSC-derived NK cells from two differentiation strategies reveals distinct signatures and cytotoxic activities. *Frontiers in immunology*. PMID 39445004 FT

Preclinical (in vivo)

- **2013** Boissel L et al. — Retargeting NK-92 cells by means of CD19- and CD20-specific chimeric antigen receptors compares favorably with antibody-dependent cellular cytotoxicity. *Oncoimmunology*. PMID 24404423 FT
- **2020** Cichocki F et al. — iPSC-derived NK cells maintain high cytotoxicity and enhance in vivo tumor control in concert with T cells and anti-PD-1 therapy. *Science translational medicine*. PMID 33148626
- **2021** Myers JA et al. — Exploring the NK cell platform for cancer immunotherapy. *Nature reviews. Clinical oncology*. PMID 32934330
- **2024** Kong D et al. — CD19 CAR-expressing iPSC-derived NK cells effectively enhance migration and cytotoxicity into glioblastoma by targeting to the pericytes in tumor microenvironment. *Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie*. PMID 38508081

Mechanistic / in vitro

- **1994** Gong JH et al. — Characterization of a human cell line (NK-92) with phenotypical and functional characteristics of activated natural killer cells. *Leukemia*. PMID 8152260
- **2010** Cooley S et al. — Donor selection for natural killer cell receptor genes leads to superior survival after unrelated transplantation for acute myelogenous leukemia. *Blood*. PMID 20581313
- **2024** Thangaraj JL et al. — Disruption of TGF- β signaling pathway is required to mediate effective killing of hepatocellular carcinoma by human iPSC-derived NK cells. *Cell stem cell*. PMID 38986609

Reviews

- **2016** Klingemann H et al. — Natural Killer Cells for Immunotherapy - Advantages of the NK-92 Cell Line over Blood NK Cells. *Frontiers in immunology*. PMID 27014270 FT
- **2021** Heipertz EL et al. — Current Perspectives on "Off-The-Shelf" Allogeneic NK and CAR-NK Cell Therapies. *Frontiers in immunology*. PMID 34925314 FT
- **2023** Berrien-Elliott MM et al. — Allogeneic natural killer cell therapy. *Blood*. PMID 36416736 FT

CAR-NK & receptor engineering, gene delivery

32 papers · 2013–2025 · 22 with culture/manufacturing content

Clinical trials

- **2018** Murad JM et al. — Manufacturing development and clinical production of NKG2D chimeric antigen receptor-expressing T cells for autologous adoptive cell therapy. *Cytotherapy*. [PMID 30180944](#)
- **2019** Burger MC et al. — CAR-Engineered NK Cells for the Treatment of Glioblastoma: Turning Innate Effectors Into Precision Tools for Cancer Immunotherapy. *Frontiers in immunology*. [PMID 31798595](#) FT
- **2021** Sun M et al. — Novel two-chain structure utilizing KIRS2/DAP12 domain improves the safety and efficacy of CAR-T cells in adults with r/r B-ALL. *Molecular therapy oncolytics*. [PMID 34703879](#) FT

Human translational

- **2023** Powers SB et al. — Differential Expression of LLT1, SLAM Receptors CS1 and 2B4 and NCR Receptors NKp46 and NKp30 in Pediatric Acute Lymphoblastic Leukemia (ALL). *International journal of molecular sciences*. [PMID 36835271](#) FT

Process / manufacturing methods

- **2019** Ingegnere T et al. — Human CAR NK Cells: A New Non-viral Method Allowing High Efficient Transfection and Strong Tumor Cell Killing. *Frontiers in immunology*. [PMID 31114587](#) FT
- **2019** Müller S et al. — High Cytotoxic Efficiency of Lentivirally and Alpharetrovirally Engineered CD19-Specific Chimeric Antigen Receptor Natural Killer Cells Against Acute Lymphoblastic Leukemia. *Frontiers in immunology*. [PMID 32117200](#) FT
- **2020** Wang X et al. — Inducible MyD88/CD40 synergizes with IL-15 to enhance antitumor efficacy of CAR-NK cells. *Blood advances*. [PMID 32384544](#)

Preclinical (in vivo)

- **2013** Chang YH et al. — A chimeric receptor with NKG2D specificity enhances natural killer cell activation and killing of tumor cells. *Cancer research*. [PMID 23302231](#)
- **2014** Chu J et al. — CS1-specific chimeric antigen receptor (CAR)-engineered natural killer cells enhance in vitro and in vivo antitumor activity against human multiple myeloma. *Leukemia*. [PMID 24067492](#)
- **2015** Schönfeld K et al. — Selective inhibition of tumor growth by clonal NK cells expressing an ErbB2/HER2-specific chimeric antigen receptor. *Molecular therapy : the journal of the American Society of Gene Therapy*. [PMID 25373520](#)
- **2015** Han J et al. — CAR-Engineered NK Cells Targeting Wild-Type EGFR and EGFRvIII Enhance Killing of Glioblastoma and Patient-Derived Glioblastoma Stem Cells. *Scientific reports*. [PMID 26155832](#) FT
- **2018** Li Y et al. — Human iPSC-Derived Natural Killer Cells Engineered with Chimeric Antigen Receptors Enhance Anti-tumor Activity. *Cell stem cell*. [PMID 30082067](#)

- **2021** Daher M et al. — Targeting a cytokine checkpoint enhances the fitness of armored cord blood CAR-NK cells. *Blood*. PMID 32902645
- **2021** Christodoulou I et al. — Engineering CAR-NK cells to secrete IL-15 sustains their anti-AML functionality but is associated with systemic toxicities. *Journal for immunotherapy of cancer*. PMID 34896980 FT
- **2021** Ma R et al. — An Oncolytic Virus Expressing IL15/IL15R α Combined with Off-the-Shelf EGFR-CAR NK Cells Targets Glioblastoma. *Cancer research*. PMID 34006525
- **2022** Li Y et al. — KIR-based inhibitory CARs overcome CAR-NK cell trogocytosis-mediated fratricide and tumor escape. *Nature medicine*. PMID 36175679
- **2022** Da Y et al. — STING agonist cGAMP enhances anti-tumor activity of CAR-NK cells against pancreatic cancer. *Oncoimmunology*. PMID 35371622 FT
- **2023** Li L et al. — Loss of metabolic fitness drives tumor resistance after CAR-NK cell therapy and can be overcome by cytokine engineering. *Science advances*. PMID 37494448 FT
- **2024** Acharya S et al. — CD28 Costimulation Augments CAR Signaling in NK Cells via the LCK/CD3 ζ /ZAP70 Signaling Axis. *Cancer discovery*. PMID 38900051
- **2024** Albinger N et al. — Manufacturing of primary CAR-NK cells in an automated system for the treatment of acute myeloid leukemia. *Bone marrow transplantation*. PMID 38253870 FT
- **2025** Luo J et al. — Neoleukin-2/15-armored CAR-NK cells sustain superior therapeutic efficacy in solid tumors via c-Myc/NRF1 activation. *Signal transduction and targeted therapy*. PMID 40025022 FT
- **2025** Peng L et al. — In vivo AAV-SB-CRISPR screens of tumor-infiltrating primary NK cells identify genetic checkpoints of CAR-NK therapy. *Nature biotechnology*. PMID 38918616
- **2025** Yang S et al. — Non-pathogenic *E. coli* displaying decoy-resistant IL18 mutein boosts anti-tumor and CAR NK cell responses. *Nature biotechnology*. PMID 39367093

Mechanistic / in vitro

- **2022** Gurney M et al. — CD38 knockout natural killer cells expressing an affinity optimized CD38 chimeric antigen receptor successfully target acute myeloid leukemia with reduced effector cell fratricide. *Haematologica*. PMID 33375774 FT
- **2023** Dagher OK et al. — Forks in the road for CAR T and CAR NK cell cancer therapies. *Nature immunology*. PMID 38012406

Reviews

- **2018** Mehta RS et al. — Chimeric Antigen Receptor Expressing Natural Killer Cells for the Immunotherapy of Cancer. *Frontiers in immunology*. PMID 29497427 FT
- **2020** Xie G et al. — CAR-NK cells: A promising cellular immunotherapy for cancer. *EBioMedicine*. PMID 32853984 FT
- **2021** Gong Y et al. — Chimeric antigen receptor natural killer (CAR-NK) cell design and engineering for cancer therapy. *Journal of hematology & oncology*. PMID 33933160 FT
- **2022** Pan K et al. — CAR race to cancer immunotherapy: from CAR T, CAR NK to CAR macrophage therapy. *Journal of experimental & clinical cancer research : CR*. PMID 35361234 FT

- **2024** Wang W et al. — Breakthrough of solid tumor treatment: CAR-NK immunotherapy. Cell death discovery. [PMID 38245520](#) FT
- **2025** Balkhi S et al. — CAR-NK cell therapy: promise and challenges in solid tumors. Frontiers in immunology. [PMID 40260240](#) FT
- **2025** Jørgensen LV et al. — The clinical landscape of CAR NK cells. Experimental hematology & oncology. [PMID 40149002](#) FT

NK-cell engagers: BiKE, TriKE, NKp46/CD16 engagers

20 papers · 2013–2026 · 14 with culture/manufacturing content

Human translational

- **2021** Kerbaux LN et al. — Combining AFM13, a Bispecific CD30/CD16 Antibody, with Cytokine-Activated Blood and Cord Blood-Derived NK Cells Facilitates CAR-like Responses Against CD30+ Malignancies. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 33986022](#)
- **2021** Reusing SB et al. — CD16xCD33 Bispecific Killer Cell Engager (BiKE) as potential immunotherapeutic in pediatric patients with AML and biphenotypic ALL. Cancer immunology, immunotherapy : CII. [PMID 34398302](#) FT

Process / manufacturing methods

- **2016** Felices M et al. — Generation of BiKEs and TriKEs to Improve NK Cell-Mediated Targeting of Tumor Cells. Methods in molecular biology (Clifton, N.J.). [PMID 27177679](#)
- **2019** Felices M et al. — Novel CD19-targeted TriKE restores NK cell function and proliferative capacity in CLL. Blood advances. [PMID 30890546](#)
- **2025** Cheng Y et al. — An IL-15-modified NKp30×HER2 trispecific NK cell engager enhances NK cell activation and tumor cell killing. Journal of leukocyte biology. [PMID 40694640](#)

Preclinical (in vivo)

- **2016** Vallera DA et al. — IL15 Trispecific Killer Engagers (TriKE) Make Natural Killer Cells Specific to CD33+ Targets While Also Inducing Persistence, In Vivo Expansion, and Enhanced Function. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 26847056](#)
- **2021** Arvindam US et al. — A trispecific killer engager molecule against CLEC12A effectively induces NK-cell mediated killing of AML cells. Leukemia. [PMID 33097838](#) FT
- **2023** Gauthier L et al. — Control of acute myeloid leukemia by a trifunctional NKp46-CD16a-NK cell engager targeting CD123. Nature biotechnology. [PMID 36635380](#) FT
- **2023** Bordoloi D et al. — Siglec-7 glyco-immune binding mAbs or NK cell engager biologics induce potent antitumor immunity against ovarian cancers. Science advances. [PMID 37910620](#) FT
- **2024** Demaria O et al. — A tetraspecific engager armed with a non-alpha IL-2 variant harnesses natural killer cells against B cell non-Hodgkin lymphoma. Science immunology. [PMID 39546590](#)

- **2025** Khaw MJ et al. — Novel trisppecific killer engager targeting B7-H3 enhances natural killer cell antitumor activity against head and neck cancer. Journal for immunotherapy of cancer. [PMID 40707133](#) FT
- **2026** Fan Y et al. — Bispecific killer cell engager-secreting CAR-T cells redirect natural killer specificity to enhance antitumour responses. Nature biomedical engineering. [PMID 40659834](#)

Mechanistic / in vitro

- **2013** Wiernik A et al. — Targeting natural killer cells to acute myeloid leukemia in vitro with a CD16 x 33 bispecific killer cell engager and ADAM17 inhibition. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 23690482](#)
- **2014** Gleason MK et al. — CD16xCD33 bispecific killer cell engager (BiKE) activates NK cells against primary MDS and MDSC CD33+ targets. Blood. [PMID 24652987](#)
- **2015** Wu J et al. — AFM13: a first-in-class tetravalent bispecific anti-CD30/CD16A antibody for NK cell-mediated immunotherapy. Journal of hematology & oncology. [PMID 26231785](#) FT
- **2022** Nikkhoi SK et al. — Bispecific killer cell engager with high affinity and specificity toward CD16a on NK cells for cancer immunotherapy. Frontiers in immunology. [PMID 36685519](#) FT
- **2025** Zambarda C et al. — CD16A Shedding Regulates Innate Cell Engager-Induced Serial Killing by Natural Killer Cells. European journal of immunology. [PMID 41116262](#) FT

Reviews

- **2019** Hodgins JJ et al. — Killers 2.0: NK cell therapies at the forefront of cancer control. The Journal of clinical investigation. [PMID 31478911](#)
- **2023** Zhang M et al. — Natural Killer Cell Engagers (NKCEs): a new frontier in cancer immunotherapy. Frontiers in immunology. [PMID 37638058](#) FT
- **2024** Shin MH et al. — Current Developments in NK Cell Engagers for Cancer Immunotherapy: Focus on CD16A and NKp46. Immune network. [PMID 39513028](#) FT

Gene editing: checkpoint and brake knockouts

24 papers · 2014–2025 · 18 with culture/manufacturing content

Human translational

- **2023** Bi J et al. — TIPE2 deletion improves the therapeutic potential of adoptively transferred NK cells. Journal for immunotherapy of cancer. [PMID 36725083](#) FT
- **2024** Xu Y et al. — Enhanced NK cell activation via eEF2K-mediated potentiation of the cGAS-STING pathway in hepatocellular carcinoma. International immunopharmacology. [PMID 38320351](#)

Process / manufacturing methods

- **2021** Huang RS et al. — A robust platform for expansion and genome editing of primary human natural killer cells. The Journal of experimental medicine. [PMID 33433623](#) FT
- **2021** Morimoto T et al. — CRISPR-Cas9-Mediated TIM3 Knockout in Human Natural Killer Cells Enhances Growth Inhibitory Effects on Human Glioma Cells. International journal of molecular sciences. [PMID 33800561](#) FT
- **2022** Bexte T et al. — CRISPR-Cas9 based gene editing of the immune checkpoint NKG2A enhances NK cell mediated cytotoxicity against multiple myeloma. Oncoimmunology. [PMID 35694192](#) FT

Preclinical (in vivo)

- **2014** Paolino M et al. — The E3 ligase Cbl-b and TAM receptors regulate cancer metastasis via natural killer cells. Nature. [PMID 24553136](#)
- **2017** Tang PM et al. — Smad3 promotes cancer progression by inhibiting E4BP4-mediated NK cell development. Nature communications. [PMID 28262747](#) FT
- **2020** Zhu H et al. — Metabolic Reprograming via Deletion of CISH in Human iPSC-Derived NK Cells Promotes In Vivo Persistence and Enhances Anti-tumor Activity. Cell stem cell. [PMID 32531207](#)
- **2020** Naeimi Kararoudi M et al. — CD38 deletion of human primary NK cells eliminates daratumumab-induced fratricide and boosts their effector activity. Blood. [PMID 32603414](#)
- **2020** Huang RS et al. — Enhanced NK-92 Cytotoxicity by CRISPR Genome Engineering Using Cas9 Ribonucleoproteins. Frontiers in immunology. [PMID 32528479](#) FT
- **2020** Hoerster K et al. — HLA Class I Knockout Converts Allogeneic Primary NK Cells Into Suitable Effectors for "Off-the-Shelf" Immunotherapy. Frontiers in immunology. [PMID 33584651](#) FT
- **2022** Bernareggi D et al. — CHMP2A regulates tumor sensitivity to natural killer cell-mediated cytotoxicity. Nature communications. [PMID 35393416](#) FT
- **2022** Bernard PL et al. — Targeting CISH enhances natural cytotoxicity receptor signaling and reduces NK cell exhaustion to improve solid tumor immunity. Journal for immunotherapy of cancer. [PMID 35589278](#) FT
- **2022** Clara JA et al. — High-affinity CD16 integration into a CRISPR/Cas9-edited CD38 locus augments CD38-directed antitumor activity of primary human natural killer cells. Journal for immunotherapy of cancer. [PMID 35135865](#) FT
- **2023** Mac Donald A et al. — KLRC1 knockout overcomes HLA-E-mediated inhibition and improves NK cell antitumor activity against solid tumors. Frontiers in immunology. [PMID 37675109](#) FT
- **2024** Bexte T et al. — CRISPR/Cas9 editing of NKG2A improves the efficacy of primary CD33-directed chimeric antigen receptor natural killer cells. Nature communications. [PMID 39349459](#) FT
- **2024** Reggiani F et al. — BET inhibitors drive Natural Killer activation in non-small cell lung cancer via BRD4 and SMAD3. Nature communications. [PMID 38519469](#) FT
- **2024** Sun X et al. — Deletion of the mRNA endonuclease Regnase-1 promotes NK cell anti-tumor activity via OCT2-dependent transcription of Ifng. Immunity. [PMID 38821052](#)

- **2024** Matson AW et al. — Enhanced IL-15-mediated NK cell activation and proliferation by an ADAM17 function-blocking antibody involves CD16A, CD137, and accessory cells. Journal for immunotherapy of cancer. [PMID 39053944](#) FT
- **2025** Biederstädt A et al. — Genome-wide CRISPR screens identify critical targets to enhance CAR-NK cell antitumor potency. Cancer cell. [PMID 40845844](#)

Mechanistic / in vitro

- **2020** Pomeroy EJ et al. — A Genetically Engineered Primary Human Natural Killer Cell Platform for Cancer Immunotherapy. Molecular therapy : the journal of the American Society of Gene Therapy. [PMID 31704085](#)
- **2025** Rea A et al. — Enhancing human NK cell antitumor function by knocking out SMAD4 to counteract TGFβ and activin A suppression. Nature immunology. [PMID 40119192](#) FT

Reviews

- **2019** Afolabi LO et al. — Genetic reprogramming for NK cell cancer immunotherapy with CRISPR/Cas9. Immunology. [PMID 31315144](#)
- **2025** Fazeli A et al. — Engineering with care: safety assessment platforms for CRISPR-modified natural killer cells. Frontiers in immunology. [PMID 41425600](#) FT

Cytokine support & armouring (IL-15, IL-21, IL-12/15/18, IL-18)

24 papers · 1993–2025 · 18 with culture/manufacturing content

Clinical trials

- **2009** Viaud S et al. — Dendritic cell-derived exosomes promote natural killer cell activation and proliferation: a role for NKG2D ligands and IL-15Ralpha. PloS one. [PMID 19319200](#) FT
- **2015** Shah NN et al. — Acute GVHD in patients receiving IL-15/4-1BBL activated NK cells following T-cell-depleted stem cell transplantation. Blood. [PMID 25452614](#)
- **2016** Romee R et al. — Cytokine-induced memory-like natural killer cells exhibit enhanced responses against myeloid leukemia. Science translational medicine. [PMID 27655849](#)
- **2022** Berrien-Elliott MM et al. — Systemic IL-15 promotes allogeneic cell rejection in patients treated with natural killer cell adoptive therapy. Blood. [PMID 34797911](#)

Human translational

- **2018** Al-Attar A et al. — Human Body Composition and Immunity: Visceral Adipose Tissue Produces IL-15 and Muscle Strength Inversely Correlates with NK Cell Function in Elderly Humans. Frontiers in immunology. [PMID 29559978](#) FT
- **2024** Li Y et al. — DNMT1 inhibition improves the activity of memory-like natural killer cells by enhancing the level of autophagy. Molecular biology reports. [PMID 39704855](#) FT

Process / manufacturing methods

- **2012** Sahm C et al. — Expression of IL-15 in NK cells results in rapid enrichment and selective cytotoxicity of gene-modified effectors that carry a tumor-specific antigen receptor. *Cancer immunology, immunotherapy* : CII. [PMID 22310931](#)
- **2019** Choi YH et al. — IL-27 enhances IL-15/IL-18-mediated activation of human natural killer cells. *Journal for immunotherapy of cancer*. [PMID 31277710](#) FT
- **2019** Kweon S et al. — Expansion of Human NK Cells Using K562 Cells Expressing OX40 Ligand and Short Exposure to IL-21. *Frontiers in immunology*. [PMID 31105701](#) FT

Preclinical (in vivo)

- **1993** D'Andrea A et al. — Interleukin 10 (IL-10) inhibits human lymphocyte interferon gamma-production by suppressing natural killer cell stimulatory factor/IL-12 synthesis in accessory cells. *The Journal of experimental medicine*. [PMID 8102388](#) FT
- **1999** Kitamura H et al. — The natural killer T (NKT) cell ligand alpha-galactosylceramide demonstrates its immunopotentiating effect by inducing interleukin (IL)-12 production by dendritic cells and IL-12 receptor expression on NKT cells. *The Journal of experimental medicine*. [PMID 10190903](#) FT
- **2009** Huntington ND et al. — IL-15 trans-presentation promotes human NK cell development and differentiation in vivo. *The Journal of experimental medicine*. [PMID 19103877](#) FT
- **2012** Denman CJ et al. — Membrane-bound IL-21 promotes sustained ex vivo proliferation of human natural killer cells. *PloS one*. [PMID 22279576](#) FT
- **2012** Ni J et al. — Sustained effector function of IL-12/15/18-preactivated NK cells against established tumors. *The Journal of experimental medicine*. [PMID 23209317](#) FT
- **2018** Liu E et al. — Cord blood NK cells engineered to express IL-15 and a CD19-targeted CAR show long-term persistence and potent antitumor activity. *Leukemia*. [PMID 28725044](#)
- **2024** Van den Eynde A et al. — IL-15-secreting CAR natural killer cells directed toward the pan-cancer target CD70 eliminate both cancer cells and cancer-associated fibroblasts. *Journal of hematology & oncology*. [PMID 38331849](#) FT
- **2024** Shanley M et al. — Interleukin-21 engineering enhances NK cell activity against glioblastoma via CEBPD. *Cancer cell*. [PMID 39137729](#)
- **2024** Berjis A et al. — Pretreatment with IL-15 and IL-18 rescues natural killer cells from granzyme B-mediated apoptosis after cryopreservation. *Nature communications*. [PMID 38729924](#) FT
- **2025** Rafei H et al. — CREM is a regulatory checkpoint of CAR and IL-15 signalling in NK cells. *Nature*. [PMID 40468083](#) FT

Mechanistic / in vitro

- **1994** Carson WE et al. — Interleukin (IL) 15 is a novel cytokine that activates human natural killer cells via components of the IL-2 receptor. *The Journal of experimental medicine*. [PMID 7523571](#) FT
- **2012** Ndhlovu LC et al. — Tim-3 marks human natural killer cell maturation and suppresses cell-mediated cytotoxicity. *Blood*. [PMID 22383801](#)

- **2024** Kennedy PR et al. — Metabolic programs drive function of therapeutic NK cells in hypoxic tumor environments. Science advances. [PMID 39475618](#) FT

Reviews

- **2015** Carlsten M et al. — Genetic Manipulation of NK Cells for Cancer Immunotherapy: Techniques and Clinical Implications. Frontiers in immunology. [PMID 26113846](#) FT
- **2018** Abel AM et al. — Natural Killer Cells: Development, Maturation, and Clinical Utilization. Frontiers in immunology. [PMID 30150991](#) FT

Ex vivo expansion, feeders, media & GMP manufacturing

30 papers · 2009–2025 · 30 with culture/manufacturing content

Clinical trials

- **2010** Cho D et al. — Cytotoxicity of activated natural killer cells against pediatric solid tumors. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 20542985](#)
- **2014** Rölle A et al. — IL-12-producing monocytes and HLA-E control HCMV-driven NKG2C+ NK cell expansion. The Journal of clinical investigation. [PMID 25384219](#)
- **2016** Leivas A et al. — Novel treatment strategy with autologous activated and expanded natural killer cells plus anti-myeloma drugs for multiple myeloma. Oncoimmunology. [PMID 28123890](#) FT

Human translational

- **2019** Zhu H et al. — An Improved Method to Produce Clinical-Scale Natural Killer Cells from Human Pluripotent Stem Cells. Methods in molecular biology (Clifton, N.J.). [PMID 31396935](#)
- **2023** Mou J et al. — 2B4 inhibits the apoptosis of natural killer cells through phosphorylated extracellular signal-related kinase/B-cell lymphoma 2 signal pathway. Cytotherapy. [PMID 37516949](#)

Process / manufacturing methods

- **2009** Cho D et al. — Expansion and activation of natural killer cells for cancer immunotherapy. The Korean journal of laboratory medicine. [PMID 19411773](#)
- **2010** Sutlu T et al. — Clinical-grade, large-scale, feeder-free expansion of highly active human natural killer cells for adoptive immunotherapy using an automated bioreactor. Cytotherapy. [PMID 20795758](#)
- **2012** Lapteva N et al. — Large-scale ex vivo expansion and characterization of natural killer cells for clinical applications. Cytotherapy. [PMID 22900959](#)
- **2013** Knorr DA et al. — Clinical-scale derivation of natural killer cells from human pluripotent stem cells for cancer therapy. Stem cells translational medicine. [PMID 23515118](#)
- **2013** Kang L et al. — Characterization and ex vivo Expansion of Human Placenta-Derived Natural Killer Cells for Cancer Immunotherapy. Frontiers in immunology. [PMID 23641243](#) FT

- **2015** Huijskens MJ et al. — Ascorbic acid promotes proliferation of natural killer cell populations in culture systems applicable for natural killer cell therapy. *Cytotherapy*. PMID 25747742
- **2017** Klöß S et al. — Optimization of Human NK Cell Manufacturing: Fully Automated Separation, Improved Ex Vivo Expansion Using IL-21 with Autologous Feeder Cells, and Generation of Anti-CD123-CAR-Expressing Effector Cells. *Human gene therapy*. PMID 28810809
- **2019** Fernández L et al. — GMP-Compliant Manufacturing of NKG2D CAR Memory T Cells Using CliniMACS Prodigy. *Frontiers in immunology*. PMID 31649672 FT
- **2020** Reina-Ortiz C et al. — Expanded NK cells from umbilical cord blood and adult peripheral blood combined with daratumumab are effective against tumor cells from multiple myeloma patients. *Oncoimmunology*. PMID 33457074 FT
- **2024** Wang X et al. — Scalable process development of NK and CAR-NK expansion in a closed bioreactor. *Frontiers in immunology*. PMID 39114666 FT
- **2025** Tran TBT et al. — In Vitro Expansion and Transduction of Primary NK Cells Using Feeder Cells Expressing Costimulatory Molecules and IL-21. *Cancer science*. PMID 40325497 FT
- **2025** Kikuchi T et al. — Scalable production process development for NK cells targeting large-scale expansion. *Regenerative therapy*. PMID 40808775 FT

Preclinical (in vivo)

- **2009** Fujisaki H et al. — Expansion of highly cytotoxic human natural killer cells for cancer cell therapy. *Cancer research*. PMID 19383914
- **2010** Siegler U et al. — Good manufacturing practice-compliant cell sorting and large-scale expansion of single KIR-positive alloreactive human natural killer cells for multiple infusions to leukemia patients. *Cytotherapy*. PMID 20491532
- **2011** Somanchi SS et al. — Expansion, purification, and functional assessment of human peripheral blood NK cells. *Journal of visualized experiments : JoVE*. PMID 21339714
- **2013** Shah N et al. — Antigen presenting cell-mediated expansion of human umbilical cord blood yields log-scale expansion of natural killer cells with anti-myeloma activity. *PloS one*. PMID 24204673 FT
- **2016** Granzin M et al. — Highly efficient IL-21 and feeder cell-driven ex vivo expansion of human NK cells with therapeutic activity in a xenograft mouse model of melanoma. *Oncoimmunology*. PMID 27757317
- **2016** Kamiya T et al. — Expanded and Activated Natural Killer Cells for Immunotherapy of Hepatocellular Carcinoma. *Cancer immunology research*. PMID 27197065
- **2016** Oyer JL et al. — Natural killer cells stimulated with PM21 particles expand and biodistribute in vivo: Clinical implications for cancer treatment. *Cytotherapy*. PMID 27059202
- **2019** Ojo EO et al. — Membrane bound IL-21 based NK cell feeder cells drive robust expansion and metabolic activation of NK cells. *Scientific reports*. PMID 31624330 FT
- **2023** Sun Y et al. — Macrophage STING signaling promotes NK cell to suppress colorectal cancer liver metastasis via 4-1BBL/4-1BB co-stimulation. *Journal for immunotherapy of cancer*. PMID 36927529 FT

- **2024** Oyer JL et al. — PM21-particle stimulation augmented with cytokines enhances NK cell expansion and confers memory-like characteristics with enhanced survival. *Frontiers in immunology*. [PMID 38711506](#) FT
- **2025** Lin H et al. — Optimizing the procedure for manufacturing clinical-grade genetically manipulated natural killer cells for adoptive immunotherapy. *Cytotherapy*. [PMID 39570247](#)

Reviews

- **2022** Gurney M et al. — Feeder Cells at the Interface of Natural Killer Cell Activation, Expansion and Gene Editing. *Frontiers in immunology*. [PMID 35222382](#) FT
- **2025** Lee J et al. — Recent advances in feeder-free NK cell expansion as a future direction of adoptive cell therapy. *Frontiers in immunology*. [PMID 41089682](#) FT

Cryopreservation, potency assays & product quality

15 papers · 2015–2026 · 15 with culture/manufacturing content

Process / manufacturing methods

- **2015** Vasu S et al. — A novel method to expand large numbers of CD56(+) natural killer cells from a minute fraction of selectively accessed cryopreserved cord blood for immunotherapy after transplantation. *Cytotherapy*. [PMID 26432560](#)
- **2018** Nham T et al. — Ex Vivo-expanded Natural Killer Cells Derived From Long-term Cryopreserved Cord Blood are Cytotoxic Against Primary Breast Cancer Cells. *Journal of immunotherapy (Hagerstown, Md. : 1997)*. [PMID 29189387](#)
- **2020** Yao X et al. — Nanoparticle-Mediated Intracellular Protection of Natural Killer Cells Avoids Cryoinjury and Retains Potent Antitumor Functions. *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*. [PMID 32382476](#) FT
- **2024** Lee S et al. — Successful expansion and cryopreservation of human natural killer cell line NK-92 for clinical manufacturing. *PloS one*. [PMID 38394177](#) FT
- **2025** Joules A et al. — Probing mechanisms of cryopreservation damage to natural killer cells. *Cytotherapy*. [PMID 39918490](#)
- **2025** Zhou X et al. — Restoring NK Cell Cytotoxicity Post-Cryopreservation via Synthetic Cells. *Advanced science (Weinheim, Baden-Wurttemberg, Germany)*. [PMID 40966444](#) FT
- **2025** Kikuchi T et al. — Evaluation of several type of chemical cryoprotectants and combination towards cryopreservation of NK cell. *Cytotechnology*. [PMID 41208818](#)
- **2026** Joules A et al. — Natural killer cell cryopreservation: Influence of composition, temperature, and granules on post-thaw survival and function. *Molecular therapy. Advances*. [PMID 42564981](#) FT
- **2026** Liu Y et al. — Induction of stress granules alleviates programmed cell death induced by lysosomal damage during NK cell cryopreservation. *Cell death discovery*. [PMID 42098076](#) FT

Mechanistic / in vitro

- **2016** Holubova M et al. — Cryopreserved NK cells in the treatment of haematological malignancies: preclinical study. Journal of cancer research and clinical oncology. PMID 27614454
- **2019** Mark C et al. — Cryopreservation impairs cytotoxicity and migration of NK cells in 3-D tissue: Implications for cancer immunotherapy. bioRxiv. 10.1101/812172
- **2020** Mark C et al. — Cryopreservation impairs 3-D migration and cytotoxicity of natural killer cells. Nature communications. PMID 33067467 FT
- **2020** Damodharan SN et al. — Analysis of ex vivo expanded and activated clinical-grade human NK cells after cryopreservation. Cytotherapy. PMID 32536506
- **2022** Reusch U et al. — Cryopreservation of Natural Killer Cells Pre-Complexed with Innate Cell Engagers. Antibodies (Basel, Switzerland). PMID 35225870 FT
- **2024** St-Denis-Bissonnette F et al. — Evaluation of resazurin phenoxazine dye as a highly sensitive cell viability potency assay for natural killer cell-derived extracellular vesicle-based cancer biotherapeutics. Journal of extracellular biology. PMID 39022723 FT

Metabolic fitness, hypoxia & in vivo persistence

18 papers · 2012–2025 · 11 with culture/manufacturing content

Human translational

- **2021** Slattery K et al. — TGF β drives NK cell metabolic dysfunction in human metastatic breast cancer. Journal for immunotherapy of cancer. PMID 33568351 FT
- **2024** Portale F et al. — C/EBP β -dependent autophagy inhibition hinders NK cell function in cancer. Nature communications. PMID 39609420 FT
- **2025** von Kries A et al. — Colorectal cancer organoids drive hypoxia, TGF- β , and patient-specific diversification of NK cell activation programs. Journal for immunotherapy of cancer. PMID 41360426 FT

Process / manufacturing methods

- **2013** Sarkar S et al. — Hypoxia induced impairment of NK cell cytotoxicity against multiple myeloma can be overcome by IL-2 activation of the NK cells. PloS one. PMID 23724099 FT
- **2018** Xu LJ et al. — Combined inhibition of JAK1,2/Stat3-PD-L1 signaling pathway suppresses the immune escape of castration-resistant prostate cancer to NK cells in hypoxia. Molecular medicine reports. PMID 29693186 FT
- **2020** Teng R et al. — Hypoxia Impairs NK Cell Cytotoxicity through SHP-1-Mediated Attenuation of STAT3 and ERK Signaling Pathways. Journal of immunology research. PMID 33123602 FT

Preclinical (in vivo)

- **2012** Sceneay J et al. — Primary tumor hypoxia recruits CD11b⁺/Ly6C^{med}/Ly6G⁺ immune suppressor cells and compromises NK cell cytotoxicity in the premetastatic niche. Cancer research. PMID 22751463

- **2013** Baginska J et al. — Granzyme B degradation by autophagy decreases tumor cell susceptibility to natural killer-mediated lysis under hypoxia. *Proceedings of the National Academy of Sciences of the United States of America*. PMID 24101526
- **2016** Viel S et al. — TGF- β inhibits the activation and functions of NK cells by repressing the mTOR pathway. *Science signaling*. PMID 26884601
- **2018** Michelet X et al. — Metabolic reprogramming of natural killer cells in obesity limits antitumor responses. *Nature immunology*. PMID 30420624
- **2018** Loftus RM et al. — Amino acid-dependent cMyc expression is essential for NK cell metabolic and functional responses in mice. *Nature communications*. PMID 29904050 FT
- **2022** Pan R et al. — Augmenting NK cell-based immunotherapy by targeting mitochondrial apoptosis. *Cell*. PMID 35447071
- **2024** Wu S et al. — The ω -3 Polyunsaturated Fatty Acid Docosahexaenoic Acid Enhances NK-Cell Antitumor Effector Functions. *Cancer immunology research*. PMID 38526128 FT

Mechanistic / in vitro

- **2024** Meng M et al. — mTOR Signaling Promotes Rapid m6A mRNA Methylation to Regulate NK-Cell Activation and Effector Functions. *Cancer immunology research*. PMID 38640466
- **2025** Wei C et al. — Nuclear mitochondrial acetyl-CoA acetyltransferase 1 orchestrates natural killer cell-dependent antitumor immunity in colorectal cancer. *Signal transduction and targeted therapy*. PMID 40289129 FT

Reviews

- **2019** Terrén I et al. — NK Cell Metabolism and Tumor Microenvironment. *Frontiers in immunology*. PMID 31616440 FT
- **2024** Miao L et al. — Advances in metabolic reprogramming of NK cells in the tumor microenvironment on the impact of NK therapy. *Journal of translational medicine*. PMID 38433193 FT
- **2025** Ghosh S et al. — Mitochondrial dynamics and metabolic attributes regulate function of natural killer cell and infiltration in tumor microenvironment modulating disease progression. *Biochimica et biophysica acta. Reviews on cancer*. PMID 41075850

Inhibitory receptors, checkpoints & tumour escape

18 papers · 2010–2025 · 13 with culture/manufacturing content

Clinical trials

- **2010** Benson DM et al. — The PD-1/PD-L1 axis modulates the natural killer cell versus multiple myeloma effect: a therapeutic target for CT-011, a novel monoclonal anti-PD-1 antibody. *Blood*. PMID 20460501
- **2018** André P et al. — Anti-NKG2A mAb Is a Checkpoint Inhibitor that Promotes Anti-tumor Immunity by Unleashing Both T and NK Cells. *Cell*. PMID 30503213 FT
- **2018** Vari F et al. — Immune evasion via PD-1/PD-L1 on NK cells and monocyte/macrophages is more prominent in Hodgkin lymphoma than DLBCL. *Blood*. PMID 29449276

- **2024** Xiong J et al. — Anti-metabolic agent pegaspargase plus PD-1 antibody sintilimab for first-line treatment in advanced natural killer T cell lymphoma. Signal transduction and targeted therapy. [PMID 38448403](#) FT

Human translational

- **2025** Sun Y et al. — Platelet-mediated circulating tumor cell evasion from natural killer cell killing through immune checkpoint CD155-TIGIT. Hepatology (Baltimore, Md.). [PMID 38779918](#)
- **2025** Wang J et al. — Differential impact of TIM-3 ligands on NK cell function. Journal for immunotherapy of cancer. [PMID 39773563](#) FT

Process / manufacturing methods

- **2018** Lanuza PM et al. — Activated human primary NK cells efficiently kill colorectal cancer cells in 3D spheroid cultures irrespectively of the level of PD-L1 expression. Oncoimmunology. [PMID 29632716](#)
- **2018** Xu L et al. — Inhibition of IL-6-JAK/Stat3 signaling in castration-resistant prostate cancer cells enhances the NK cell-mediated cytotoxicity via alteration of PD-L1/NKG2D ligand levels. Molecular oncology. [PMID 28865178](#) FT
- **2019** Carlsten M et al. — Bortezomib sensitizes multiple myeloma to NK cells via ER-stress-induced suppression of HLA-E and upregulation of DR5. Oncoimmunology. [PMID 30713790](#) FT

Preclinical (in vivo)

- **2018** Zhang Q et al. — Blockade of the checkpoint receptor TIGIT prevents NK cell exhaustion and elicits potent anti-tumor immunity. Nature immunology. [PMID 29915296](#)
- **2018** Hsu J et al. — Contribution of NK cells to immunotherapy mediated by PD-1/PD-L1 blockade. The Journal of clinical investigation. [PMID 30198904](#)
- **2023** Liu X et al. — Immune checkpoint HLA-E:CD94-NKG2A mediates evasion of circulating tumor cells from NK cell surveillance. Cancer cell. [PMID 36706761](#)
- **2024** Taylor BC et al. — NKG2A Is a Therapeutic Vulnerability in Immunotherapy Resistant MHC-I Heterogeneous Triple-Negative Breast Cancer. Cancer discovery. [PMID 37791898](#)
- **2025** Xie J et al. — Overexpressing natural killer group 2 member A drives natural killer cell exhaustion in relapsed acute myeloid leukemia. Signal transduction and targeted therapy. [PMID 40320412](#) FT

Mechanistic / in vitro

- **2024** Lupo KB et al. — synNotch-programmed iPSC-derived NK cells usurp TIGIT and CD73 activities for glioblastoma therapy. Nature communications. [PMID 38429294](#) FT

Reviews

- **2019** Pende D et al. — Killer Ig-Like Receptors (KIRs): Their Role in NK Cell Modulation and Developments Leading to Their Clinical Exploitation. Frontiers in immunology. [PMID 31231370](#) FT

- **2020** Khan M et al. — NK Cell-Based Immune Checkpoint Inhibition. *Frontiers in immunology*. [PMID 32117298](#) FT
- **2024** Li Y et al. — Unlocking the therapeutic potential of the NKG2A-HLA-E immune checkpoint pathway in T cells and NK cells for cancer immunotherapy. *Journal for immunotherapy of cancer*. [PMID 39486805](#) FT

ADCC, antibody and drug combinations

18 papers · 2008–2025 · 15 with culture/manufacturing content

Clinical trials

- **2017** Canter RJ et al. — Radiotherapy enhances natural killer cell cytotoxicity and localization in pre-clinical canine sarcomas and first-in-dog clinical trial. *Journal for immunotherapy of cancer*. [PMID 29254507](#) FT
- **2022** Walle T et al. — Radiotherapy orchestrates natural killer cell dependent antitumor immune responses through CXCL8. *Science advances*. [PMID 35319989](#) FT
- **2022** Nakamura A et al. — The SUMOylation inhibitor subasumstat potentiates rituximab activity by IFN1-dependent macrophage and NK cell stimulation. *Blood*. [PMID 35226739](#) FT
- **2024** Choi MG et al. — Safety and efficacy of SNK01 (autologous natural killer cells) in combination with cytotoxic chemotherapy and/or cetuximab after failure of prior tyrosine kinase inhibitor in non-small cell lung cancer: non-clinical mouse model and phase I/IIa clinical study. *Journal for immunotherapy of cancer*. [PMID 38538093](#) FT

Human translational

- **2015** Jie HB et al. — CTLA-4⁺ Regulatory T Cells Increased in Cetuximab-Treated Head and Neck Cancer Patients Suppress NK Cell Cytotoxicity and Correlate with Poor Prognosis. *Cancer research*. [PMID 25832655](#)
- **2020** Park JE et al. — Anti-tumor effects of NK cells and anti-PD-L1 antibody with antibody-dependent cellular cytotoxicity in PD-L1-positive cancer cell lines. *Journal for immunotherapy of cancer*. [PMID 32830112](#) FT
- **2024** Kumar B et al. — BATF is a major driver of NK cell epigenetic reprogramming and dysfunction in AML. *Science translational medicine*. [PMID 39259809](#)

Process / manufacturing methods

- **2008** Binyamin L et al. — Blocking NK cell inhibitory self-recognition promotes antibody-dependent cellular cytotoxicity in a model of anti-lymphoma therapy. *Journal of immunology* (Baltimore, Md. : 1950). [PMID 18424763](#)
- **2021** Gopal S et al. — 3D tumor spheroid microarray for high-throughput, high-content natural killer cell-mediated cytotoxicity. *Communications biology*. [PMID 34290356](#) FT
- **2025** Maskalenko NA et al. — The FcγRIIIA (CD16) L48-H/R Polymorphism Enhances NK Cell-Mediated Antibody-Dependent Cellular Cytotoxicity by Promoting Serial Killing. *Cancer immunology research*. [PMID 39666369](#)

Preclinical (in vivo)

- **2015** Yin J et al. — Ezh2 regulates differentiation and function of natural killer cells through histone methyltransferase activity. Proceedings of the National Academy of Sciences of the United States of America. [PMID 26668377](#)
- **2018** Weiss T et al. — NKG2D-Based CAR T Cells and Radiotherapy Exert Synergistic Efficacy in Glioblastoma. Cancer research. [PMID 29222400](#)
- **2020** Zhu H et al. — Pluripotent stem cell-derived NK cells with high-affinity noncleavable CD16a mediate improved antitumor activity. Blood. [PMID 31856277](#)
- **2024** Arora J et al. — T-cell help in the tumor microenvironment enhances rituximab-mediated NK-cell ADCC. Blood. [PMID 38457360](#)

Mechanistic / in vitro

- **2024** Liu J et al. — Epigenetic regulation of CD38/CD48 by KDM6A mediates NK cell response in multiple myeloma. Nature communications. [PMID 38355622](#) FT

Reviews

- **2013** Seidel UJ et al. — Natural killer cell mediated antibody-dependent cellular cytotoxicity in tumor immunotherapy with therapeutic antibodies. Frontiers in immunology. [PMID 23543707](#) FT
- **2015** Wang W et al. — NK Cell-Mediated Antibody-Dependent Cellular Cytotoxicity in Cancer Immunotherapy. Frontiers in immunology. [PMID 26284063](#) FT
- **2025** Vincken R et al. — ADCC: the rock band led by therapeutic antibodies, tumor and immune cells. Frontiers in immunology. [PMID 40308580](#) FT

Trafficking, homing & the solid-tumour microenvironment

18 papers · 2003–2025 · 12 with culture/manufacturing content

Clinical trials

- **2012** Somanchi SS et al. — Engineering lymph node homing of ex vivo-expanded human natural killer cells via trogocytosis of the chemokine receptor CCR7. Blood. [PMID 22498742](#)
- **2020** Okoye I et al. — Galectin-9 expression defines exhausted T cells and impaired cytotoxic NK cells in patients with virus-associated solid tumors. Journal for immunotherapy of cancer. [PMID 33310773](#) FT
- **2022** Schomer NT et al. — CCR7 expression in CD19 chimeric antigen receptor-engineered natural killer cells improves migration toward CCL19-expressing lymphoma cells and increases tumor control in mice with human lymphoma. Cytotherapy. [PMID 35400595](#)

Human translational

- **2024** Neo SY et al. — Tumor-associated NK cells drive MDSC-mediated tumor immune tolerance through the IL-6/STAT3 axis. Science translational medicine. [PMID 38748775](#)

- **2024** Fang H et al. — myCAF-derived exosomal PWAR6 accelerates CRC liver metastasis via altering glutamine availability and NK cell function in the tumor microenvironment. *Journal of hematology & oncology*. [PMID 39696364](#) FT
- **2025** Russo E et al. — Cxcr3 promotes protection from colorectal cancer liver metastasis by driving NK cell infiltration and plasticity. *The Journal of clinical investigation*. [PMID 40168086](#) FT

Process / manufacturing methods

- **2018** Huergo-Zapico L et al. — NK-cell Editing Mediates Epithelial-to-Mesenchymal Transition via Phenotypic and Proteomic Changes in Melanoma Cell Lines. *Cancer research*. [PMID 29752261](#)
- **2022** Marzagalli M et al. — A multi-organ-on-chip to recapitulate the infiltration and the cytotoxic activity of circulating NK cells in 3D matrix-based tumor model. *Frontiers in bioengineering and biotechnology*. [PMID 35957642](#) FT
- **2024** Karlsson V et al. — Elevated Galectin-3 levels in the tumor microenvironment of ovarian cancer - implication of ROS mediated suppression of NK cell antitumor response via tumor-associated neutrophils. *Frontiers in immunology*. [PMID 39759523](#) FT

Preclinical (in vivo)

- **2003** Ancuta P et al. — Fractalkine preferentially mediates arrest and migration of CD16+ monocytes. *The Journal of experimental medicine*. [PMID 12810688](#) FT
- **2018** Böttcher JP et al. — NK Cells Stimulate Recruitment of cDC1 into the Tumor Microenvironment Promoting Cancer Immune Control. *Cell*. [PMID 29429633](#) FT
- **2018** Barry KC et al. — A natural killer-dendritic cell axis defines checkpoint therapy-responsive tumor microenvironments. *Nature medicine*. [PMID 29942093](#)
- **2024** He C et al. — Vitamin B6 Competition in the Tumor Microenvironment Hampers Antitumor Functions of NK Cells. *Cancer discovery*. [PMID 37931287](#) FT
- **2025** Ben-Shmuel A et al. — Cancer-Associated Fibroblasts Serve as Decoys to Suppress NK Cell Anticancer Cytotoxicity in Breast Cancer. *Cancer discovery*. [PMID 40052789](#)
- **2025** Jin J et al. — Targeting lactylation reinforces NK cell cytotoxicity within the tumor microenvironment. *Nature immunology*. [PMID 40494934](#)

Reviews

- **2020** Shimasaki N et al. — NK cells for cancer immunotherapy. *Nature reviews. Drug discovery*. [PMID 31907401](#)
- **2022** Ran GH et al. — Natural killer cell homing and trafficking in tissues and tumors: from biology to application. *Signal transduction and targeted therapy*. [PMID 35768424](#) FT
- **2024** Peng L et al. — CAR-T and CAR-NK as cellular cancer immunotherapy for solid tumors. *Cellular & molecular immunology*. [PMID 39134804](#) FT

Clinical evidence: adoptive NK, CAR-NK, memory-like NK

28 papers · 1996–2025 · 26 with culture/manufacturing content

Clinical trials

- **2005** Miller JS et al. — Successful adoptive transfer and in vivo expansion of human haploidentical NK cells in patients with cancer. *Blood*. PMID 15632206
- **2008** Arai S et al. — Infusion of the allogeneic cell line NK-92 in patients with advanced renal cell cancer or melanoma: a phase I trial. *Cytotherapy*. PMID 18836917
- **2011** Geller MA et al. — A phase II study of allogeneic natural killer cell therapy to treat patients with recurrent ovarian and breast cancer. *Cytotherapy*. PMID 20849361
- **2011** Curti A et al. — Successful transfer of alloreactive haploidentical KIR ligand-mismatched natural killer cells after infusion in elderly high risk acute myeloid leukemia patients. *Blood*. PMID 21791425
- **2014** Bachanova V et al. — Clearance of acute myeloid leukemia by haploidentical natural killer cells is improved using IL-2 diphtheria toxin fusion protein. *Blood*. PMID 24719405
- **2015** Conlon KC et al. — Redistribution, hyperproliferation, activation of natural killer cells and CD8 T cells, and cytokine production during first-in-human clinical trial of recombinant human interleukin-15 in patients with cancer. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*. PMID 25403209
- **2015** Rothe A et al. — A phase 1 study of the bispecific anti-CD30/CD16A antibody construct AFM13 in patients with relapsed or refractory Hodgkin lymphoma. *Blood*. PMID 25887777
- **2017** Ciurea SO et al. — Phase 1 clinical trial using mbIL21 ex vivo-expanded donor-derived NK cells after haploidentical transplantation. *Blood*. PMID 28835441
- **2019** Baumeister SH et al. — Phase I Trial of Autologous CAR T Cells Targeting NKG2D Ligands in Patients with AML/MDS and Multiple Myeloma. *Cancer immunology research*. PMID 30396908
- **2019** Cooley S et al. — First-in-human trial of rhIL-15 and haploidentical natural killer cell therapy for advanced acute myeloid leukemia. *Blood advances*. PMID 31266741
- **2020** Liu E et al. — Use of CAR-Transduced Natural Killer Cells in CD19-Positive Lymphoid Tumors. *The New England journal of medicine*. PMID 32023374
- **2023** Burger MC et al. — Intracranial injection of natural killer cells engineered with a HER2-targeted chimeric antigen receptor in patients with recurrent glioblastoma. *Neuro-oncology*. PMID 37148198 FT
- **2024** Marin D et al. — Safety, efficacy and determinants of response of allogeneic CD19-specific CAR-NK cells in CD19+ B cell tumors: a phase 1/2 trial. *Nature medicine*. PMID 38238616 FT
- **2025** Nieto Y et al. — Allogeneic NK cells with a bispecific innate cell engager in refractory relapsed lymphoma: a phase 1 trial. *Nature medicine*. PMID 40186077
- **2025** Lei W et al. — Safety and feasibility of 4-1BB co-stimulated CD19-specific CAR-NK cell therapy in refractory/relapsed large B cell lymphoma: a phase 1 trial. *Nature cancer*. PMID 40251398 FT
- **2025** Ghobadi A et al. — Induced pluripotent stem-cell-derived CD19-directed chimeric antigen receptor natural killer cells in B-cell lymphoma: a phase 1, first-in-human trial. *Lancet (London, England)*. PMID 39798981

- **2025** Shapiro RM et al. — First-in-human evaluation of memory-like NK cells with an IL-15 super-agonist and CTLA-4 blockade in advanced head and neck cancer. *Journal of hematology & oncology*. [PMID 39948608](#) FT
- **2025** Huang R et al. — Safety and efficacy of CD33-targeted CAR-NK cell therapy for relapsed/refractory AML: preclinical evaluation and phase I trial. *Experimental hematology & oncology*. [PMID 39748428](#) FT

Human translational

- **2016** Stringaris K et al. — KIR gene haplotype: an independent predictor of clinical outcome in MDS patients. *Blood*. [PMID 27760759](#)
- **2025** Korst CLBM et al. — The bone marrow NK-cell profile predicts MRD negativity in patients with multiple myeloma treated with daratumumab-based therapy. *Blood*. [PMID 40019438](#)

Process / manufacturing methods

- **1996** Pierson BA et al. — CD56+bright and CD56+dim natural killer cells in patients with chronic myelogenous leukemia progressively decrease in number, respond less to stimuli that recruit clonogenic natural killer cells, and exhibit decreased proliferation on a per cell basis. *Blood*. [PMID 8822949](#)
- **2009** Berg M et al. — Clinical-grade ex vivo-expanded human natural killer cells up-regulate activating receptors and death receptor ligands and have enhanced cytolytic activity against tumor cells. *Cytotherapy*. [PMID 19308771](#)
- **2011** Kreutzman A et al. — Expansion of highly differentiated CD8+ T-cells or NK-cells in patients treated with dasatinib is associated with cytomegalovirus reactivation. *Leukemia*. [PMID 21647156](#)

Preclinical (in vivo)

- **2002** Ruggeri L et al. — Effectiveness of donor natural killer cell alloreactivity in mismatched hematopoietic transplants. *Science (New York, N.Y.)*. [PMID 11896281](#)
- **2005** Ghiringhelli F et al. — CD4+CD25+ regulatory T cells inhibit natural killer cell functions in a transforming growth factor-beta-dependent manner. *The Journal of experimental medicine*. [PMID 16230475](#) FT
- **2009** Hoechst B et al. — Myeloid derived suppressor cells inhibit natural killer cells in patients with hepatocellular carcinoma via the NKp30 receptor. *Hepatology (Baltimore, Md.)*. [PMID 19551844](#)

Reviews

- **2011** Moretta L et al. — Killer Ig-like receptor-mediated control of natural killer cell alloreactivity in haploidentical hematopoietic stem cell transplantation. *Blood*. [PMID 20889923](#)
- **2022** Lamers-Kok N et al. — Natural killer cells in clinical development as non-engineered, engineered, and combination therapies. *Journal of hematology & oncology*. [PMID 36348457](#) FT